



Advancements in Machine Learning for Brain Tumor Classification and Diagnosis: A Comprehensive Review of Challenges and Future Directions

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Abstract

Brain tumors (BrTs), as one of the most complex and challenging intracranial neoplasms, are of particular importance in medical science due to their aggressive growth and resistance to treatment. Early detection of tumors and their type in the early stages is critical. Machine learning (ML) techniques, as one of the most advanced artificial intelligence (AI) technologies, have considerable potential for the development of BrT therapy and detection processes. However, there are several challenges in utilizing ML in the medical detection and therapy of BrTs. Limitations related to insufficient training data, data heterogeneity, interpretability of deep learning (DL) models, ethical challenges, and patient data privacy are among the key obstacles to the widespread implementation of these techniques in clinical settings. This work provides a comprehensive review of the applications of ML in diagnosing and treating BrTs. This paper systematically reviews the methods of DL, federated learning (FL), reinforcement learning (RL), and hybrid models used in detecting and treating BrTs by providing a detailed taxonomy and a comprehensive classification of these technologies. The main objective of this research is to evaluate the effectiveness of ML systems in improving diagnostic and therapeutic processes. Analysis of the application of these techniques in the therapy and detection of BrTs shows that the main focus of these studies is on precise tumor detection (PTD) with 32% and efficient computational processing (ECP) with 22%. These findings indicate that improving PTD and ECP should be the main priority in developing ML systems for diagnosing and treating BrTs. The results of this research emphasize that using ML can lead to increased accuracy in medical image analysis, reduced human errors, improved performance of predictive models, and optimized clinical decision-making. Finally, this study paves the way for the effective integration of ML techniques in identifying and curing BrTs By evaluating existing issues and proposing enhancement strategies.

1 Introduction

BrTs, as one of the most intricate and demanding intracranial neoplasms, have always attracted the attention of researchers and medical professionals due to their invasive nature and destructive effects on neurological function. These tumors, which may originate from brain parenchymal cells such as astrocytes, oligodendrocytes, and ependymocytes

or non-native elements such as meningeal and vascular cells, are classified based on the degree of cellular differentiation, proliferation rate, and degree of invasion into adjacent tissues [1, 2]. Genetic factors, cellular mutations, environmental influences, and complex molecular mechanisms drive BrT development by promoting uncontrolled cell growth, enhancing cell death resistance, and enabling immune mechanism evasion. These factors, combined with the brain's complex structure and the blood-brain barrier's limitations, have created significant challenges in identifying and managing this illness [3]. ML is a branch of AI that designs algorithms that analyze patterns and extract hidden features from large amounts of data to enable prediction, automated decision-making, and process optimization. ML models include supervised, unsupervised, and RL methods in areas ranging from natural language processing to image

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and numerical data analysis. This technology has improved disease diagnosis accuracy, personalized treatment, disease progression prediction, and reduced medical costs in the health sector. Analyzing genetic data and medical records using ML algorithms enables early detection of complex diseases and provides treatments tailored to each patient's characteristics. Also, processing medical images with the assistance of deep neural networks (DNN) reduces human error and provides more accurate diagnoses [4, 5].

1.1 Motivation and Challenges

The therapy and detection of BrTs by traditional methods face several challenges that severely affect the efficiency and accuracy of these processes. One of the main problems is the limitations of medical imaging in accurately distinguishing normal tissues from tumor tissues, especially in the early stages of the disease [6]. In addition, the evaluation of medical images is greatly influenced by the knowledge and skill of radiologists, and the possibility of human error increases, especially in complex cases. In the treatment field, challenges such as choosing the appropriate treatment strategy based on individual patient characteristics, predicting response to treatment, and managing side effects of drugs and radiation therapy are among the significant obstacles [7, 8]. Conventional treatments, such as surgery, radiotherapy, and chemotherapy, often have significant side effects that reduce the quality of life of patients and sometimes cause treatment resistance and relapse. Implementing ML techniques in identifying and curing BrTs has overcome many of the challenges of traditional methods and improved the accuracy and efficiency of these processes. DL models, especially convolutional neural networks (CNNs), have enabled the analysis of medical images with accuracy beyond human capabilities and have enabled more accurate and earlier diagnosis of tumors by extracting complex features in MRI and CT scans. Also, RL and transfer learning models have accelerated the processing of extensive image data and optimized the interpretation of medical images by reducing dependence on labeled data. FL models have also led to the integration of clinical data and improved medical decision-making by processing hospital data in a distributed manner without violating privacy. These advancements, coupled with improved medical resource allocation and reduced human error, have facilitated the transformation of diagnostic and therapeutic systems, significantly enhancing the quality of medical care [9]. Previous studies in brain tumor diagnosis and treatment using ML have made significant progress but still face several challenges. Some research has focused on using deep neural networks (DNN) for medical image processing. Still, these methods often require large and labeled datasets that are difficult to access

in clinical settings [10]. Also, previous studies in the field of FL have shown that this method can help share information between healthcare centers without violating patient privacy. Still, issues related to model synchronization and data heterogeneity remain obstacles [11]. In addition, some review articles have focused on the role of RL in treatment optimization. Still, challenges such as computational complexity and the need for large training data have posed serious limitations [12]. This article offers a thorough overview of the use of ML techniques in treating and detecting BrTs. This paper reviews advanced technologies such as DL, transfer learning, FL, RL, and hybrid methods to present an organized framework and categorization for their use in identifying and curing BrTs. Previous studies have explored ML techniques for BrT detection and treatment, but challenges such as data scarcity, model interpretability, privacy concerns in FL, and computational complexity in RL remain unresolved. Unlike prior works focusing on specific methods, this review provides a comprehensive framework to systematically evaluate and compare DL, FL, RL, and hybrid approaches, addressing key limitations and proposing improvement strategies. The main goal of this review is to identify and analyze the strengths and weaknesses of traditional methods in the therapy and detection of BrTs. The paper also attempts to identify current research gaps and provide researchers with opportunities for improvement and development in designing targeted diagnostic and therapeutic models.

1.2 Our Contribution

This section provides a comprehensive framework for analyzing and classifying ML techniques in BrT therapy and detection. This section identifies the strengths and weaknesses of traditional methods and presents innovative solutions based on new algorithms. Our contribution to this work includes the following:

- Providing a systematic classification of ML techniques (such as deep, transfer, federated, reinforcement, and hybrid learning).
- Analyzing the strengths and weaknesses of traditional methods in the therapy and detection of BrTs.
- Investigating and comparing the effectiveness of new models in terms of precision, efficiency, and minimizing human mistakes.
- Identifying existing research gaps and providing suggestions for improving diagnostic and treatment algorithms.
- Providing solutions for integrating clinical data using distributed models to increase the quality of medical care.

- Providing a comprehensive and reference view for researchers to guide future research in this field.

1.3 Organization of the Article

This study presents its content in an integrated manner. In Sect. 2, the authors provide a comprehensive review of the history of BrTs, various aspects of the disease, and traditional diagnostic and therapeutic methods, and they introduce ML technologies along with their role in improving clinical processes. In Sect. 3, the authors review and evaluate relevant research articles and compare the performance of ML techniques in identifying and curing BrTs. In Sect. 4, the study details the research methodology by outlining the steps researchers followed for selecting and evaluating articles. In Sect. 5, the authors introduce a systematic classification system for ML applications in identifying and curing BrTs and briefly review the characteristics and differences of each method. In Sect. 6, the study provides an in-depth analysis of key comparative factors through critical discussions. In Sect. 7, the authors offer suggestions and recommendations for optimally utilizing these techniques to enhance diagnostic and therapeutic processes for BrTs. Finally, in Sect. 8, the study concludes and summarizes the material presented in this paper.

2 Background

This section explores the application of ML strategies in identifying and curing BrTs, emphasizing key aspects of these techniques in improving early detection, personalized treatment strategies, and enhancing patient clinical outcomes.

2.1 BrTs Cancer

BrTs are a heterogeneous group of intra-skull neoplasms classified by tissue origin, degree of cell distinction, proliferation speed, and invasion into adjacent structures. These neoplasms can originate from brain parenchymal cells, including astrocytes, oligodendrocytes, ependymocytes, or non-parenchymal elements such as meningeal, vascular, or pituitary-derived cells [13, 14]. Early tumors such as high-grade glioblastoma (multiform glioblastoma) have high invasive potential, extensive angiogenesis, multiple mutations on intra-soluble messenger paths, and resistance to conventional treatments. In contrast, cerebral metastatic tumors, which mainly derive from primary carcinomas of the lung, breast, malignant melanoma, or carcinoma, are spread to the brain through hematogenic pathways and create neurotic and neurological symptoms [15–17]. Diagnosis

of these neoplasms is possible by multimodal imaging, including MRIs with advanced sequences (SWI, DWI, MRS, FMRI) and positron-based tomography (PET) based on dedicated detectors such as ^{18}F -Fet and ^{11}C -Met. A surgical biopsy or resection is essential for histopathological examination and immunohistochemical tests (such as IDH1, ATRX, MGMT, and P53). Therapeutic strategies include multidimensional approaches such as complete macroscopic resection, adaptive radiotherapy (IMRT, SRS), alcohol-based chemotherapy (Temozolomide, lomustine), and targeted therapies such as VEGF inhibitors, or Bortezomib. In addition, emerging methods such as immunotherapy with checkpoint inhibitors (PD-1/PD-L1 Inhibitors), personalized neo-based vaccines and gene therapy investigated. Therapeutic decision-making depends on factors such as the Karnofsky functional index, molecular biomarkers, patient age, and tumor penetration in the functional areas of the brain [18, 19]. It requires interdisciplinary management by neurooncologists, neurosurgeons, radiooncologists, and cancer specialists. The formation of the BrT is shown in Fig. 1.

2.2 BrTs Method Detection

Diagnosis of BrTs based on existing clinical and laboratory methods combines medical imaging techniques, histological analytics, and molecular biomarkers that directly examine the tumor's physiological, metabolic, and pathological properties. Magnetic Resonance Imaging (MRI) Multiparametric includes advanced sequences such as FLAIR, T1/T2-weighted, Diffusion-Weighted Imaging (DWI), and Perfusion-Weighted Imaging (PWI), the ability to evaluate structural properties, texture, and vascularity [20]. On the other hand, Magnetic Resonance Spectroscopy (MRS) examines the metabolic changes caused by the accumulation of choline, lactate, and reduced N-acetyl aspartate (NAA) as markers of neoplastic changes and helps differentiate high-grade gliomas [21]. Along with imaging methods, Stereotactic Biopsy, using 3D imaging guidance, facilitates suspicious tissue sampling and enables histopathological analysis based on hematoxylin and eosin (H&E). Examination of Key Protein Markers such as Ki-67 (to evaluate cellular proliferation rates), IDH1 R132H (for the differentiation of low-grade gliomas), and ATRX (to detect the pattern of mutations associated with osteosarcoma) are critical factors in the molecular classification of these tumors [22, 23]. In addition, tests of Cerebrospinal Fluid (CSF) and Liquid Biopsy are used to identify Circulating Tumor DNA (ctDNA), microRNAs (miRNAs), and MGMT Promoter Methylation as non-invasive biomarkers in prognosis and response to treatment. Positron Emission Tomography (PET) based on radioactive tracers such as FDG and

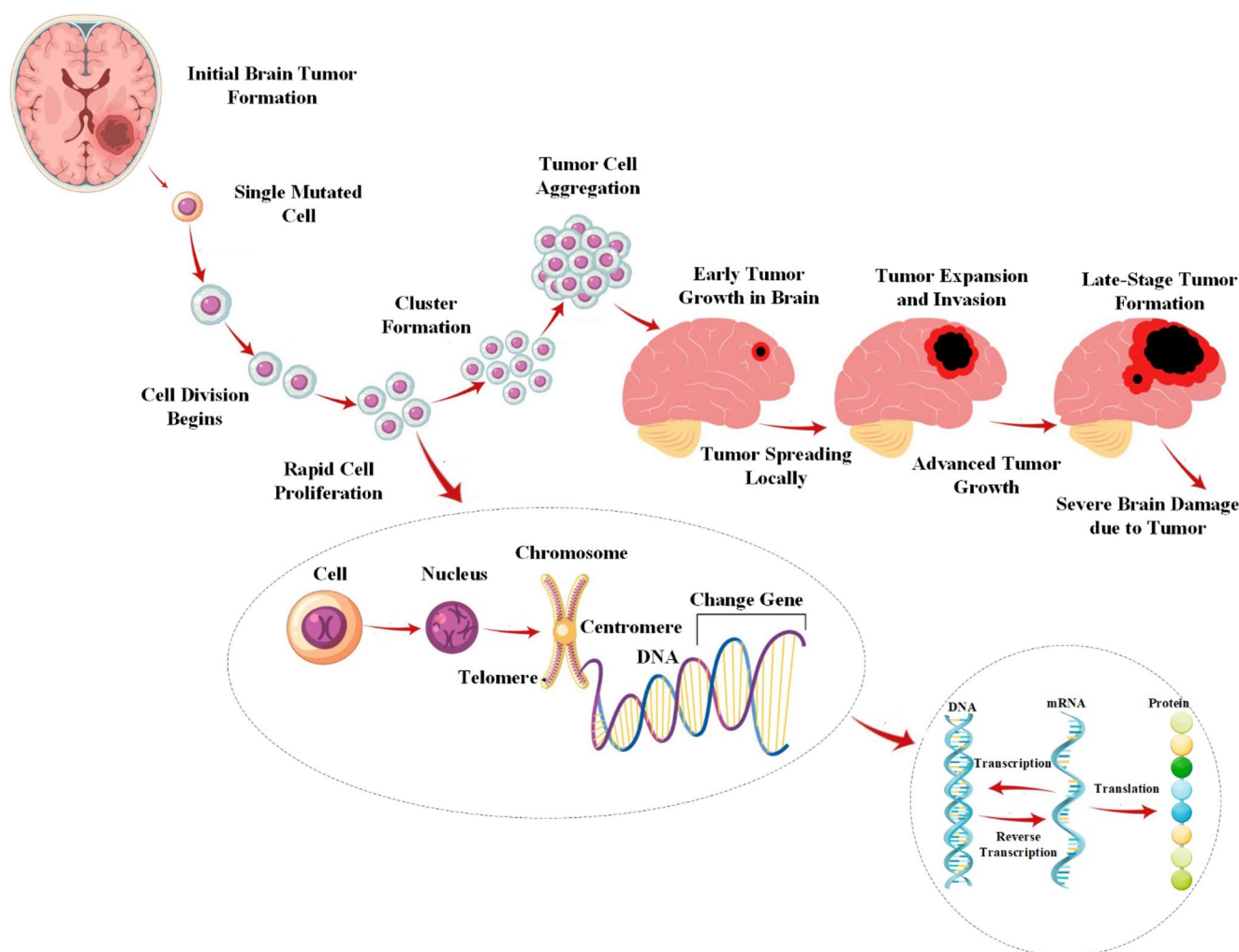


Fig. 1 Stages of BrT Development and Cellular Progression

Met-PET as a complementary tool enables the analysis of the metabolic activity of the tumor area and is essential in distinguishing between neoplastic and inflammatory lesions. By combining radiological, histopathological, and biochemical approaches, these diagnostic methods enable the comprehensive evaluation of the nature of BrTs and are based on determining individual-based therapeutic strategies [24, 25]. In Fig. 2, the important techniques for diagnosing BrTs are shown.

2.3 BrTs Method Treatment

Treatment of BrTs is a complex challenge that requires multidimensional and specialized approaches, which combine surgery, radiation therapy, chemotherapy, modern immunotherapy, gene therapy, and targeted therapies. The approach to treatment is based on the type of tumor, its position, the degree of malignancy, and the specific molecular characteristics of the cancer. Surgery is still one of the main pillars

of BrTs, especially in tumors that are accessible and may be harvested entirely in clinical or imaging. Advanced stereotactic surgery techniques and microscopic surgery allow surgeons to extract tumors with high accuracy, even when located in sensitive areas of the brain. Radiotherapy is used to treat BrTs, especially in cases where complete or safe surgery is not possible. External Beam Radiation Therapy and Stereotactic Radiosurgery, which includes Gamma Knife and CyberKnife, help to treat highly careful BrTs [26, 27]. Medications such as temozolomide and lymphokine (Lomustine) are used to reduce tumor growth and sometimes to increase the effects of radiation therapy. Chemotherapy is typically used after surgery to prevent tumor recurrence and reduce the probability of metastasis. However, because of the lack of access to the brain (due to the presence of a blood-brain barrier), chemotherapy alone has limitations. In recent years, targeted and immunotherapy treatments have become increasingly considered in treating BrTs. Immunotherapy, which significantly stimulates the

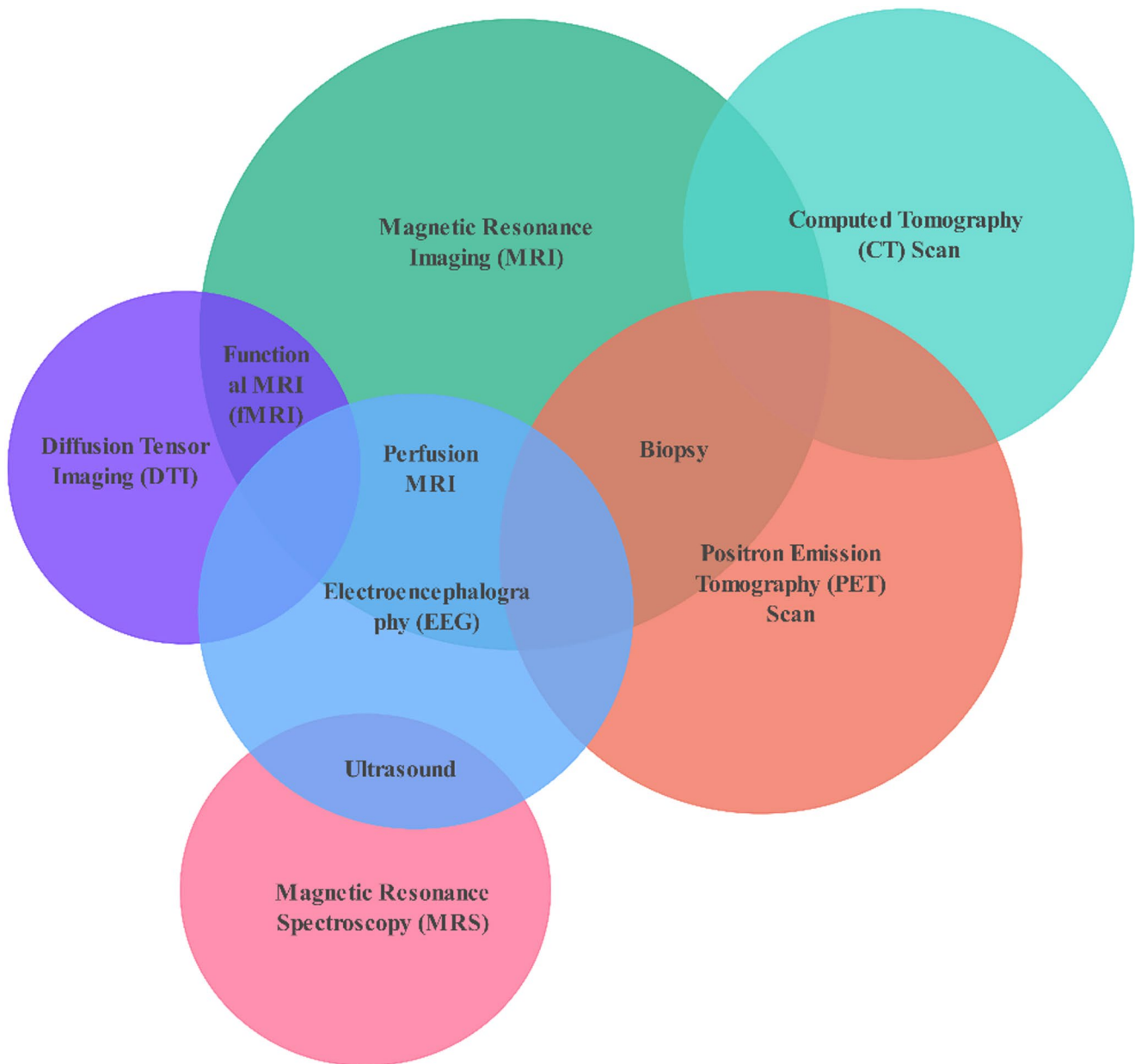


Fig. 2 Overview of Commonly Used BrT Diagnosis Methods

immune system to recognize and destroy cancerous cells, is recognized as a modern treatment with many therapeutic potentials. Targeted treatments such as monoclonal antibodies (e.g., bevacizumab, which helps inhibit tumor vessels) and kinase inhibitors can target specific signal pathways [28, 29]. Figure 3 shows essential treatment methods.

2.4 ML Technology

ML is one of the branches of AI that focuses on developing algorithms that, without explicit programming, gain the ability to learn and improve their performance through

experience [30]. This technology has become one of the key tools in data science, engineering, medicine, economics, and other specialized fields due to the growth of digital data and the need for rapid information processing and analysis. The primary purpose of learning the machine is to identify complex patterns, make precise predictions, and automate decision-making processes so that the system can optimally face new data. How to use ML depends on the type of problem, data volume, and computational resources. The overall process involves data collection and pre-processing, selection and engineering of features, selection of the right model, model training, performance evaluation,

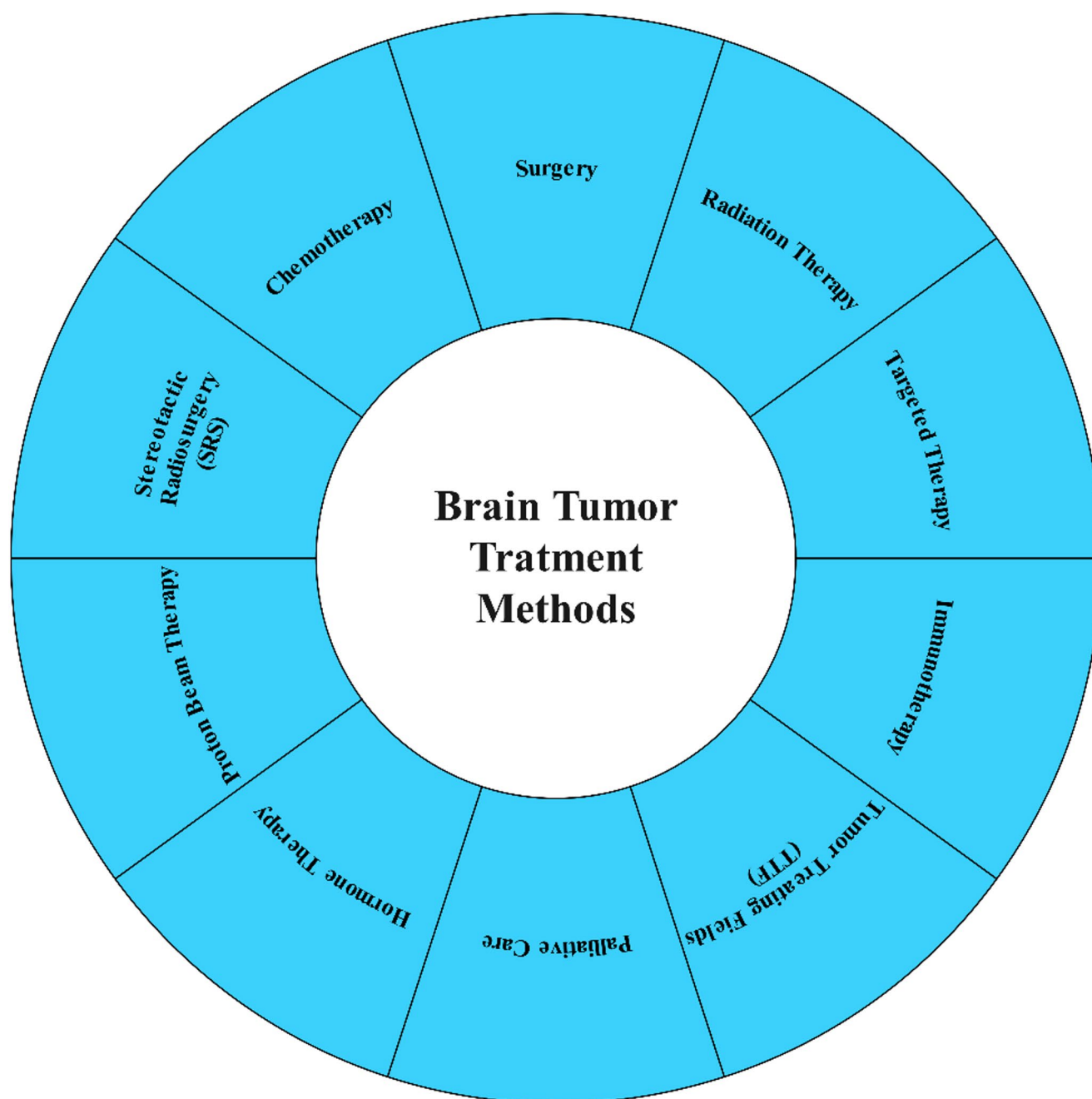


Fig. 3 Overview of commonly used BrT treatment methods

and optimization [31, 32]. Techniques such as Transfer, Federated, and self-supervised learning help develop optimal performance models in limited data conditions. Using modern processing infrastructure such as GPU and TPU, today's sophisticated models such as Deep Neural Networks (DNN) and generative models such as Transformers and Generative Adversarial Networks (GANs) have enabled enormous data and precision forecasts [33]. New techniques such as Diffusion Models, which have progressed in producing artificial data and reconstructing images and videos,

are recognized as one of the latest achievements in the field. Also, multimodal learning, capable of processing several data types, such as text, image, and sound, has dramatically improved applications such as conversational AI and video analysis. On the other hand, advances in Few-Shot Learning and Zero-Shot Learning have made it possible to generalize models in situations where educational data are limited. In addition, adaptive learning and continual learning allow models to be updated continuously in variable environments, which are essential in real-time applications such as

automation systems, anomalies, and medical data analysis [34]. Figure 4 illustrates how ML technology works. ML techniques have significantly improved the detection of BrTs, increasing accuracy and speed and automating medical image analysis.

Extracting spatial properties from medical scans allows tumor detection and separation and increases the accuracy of the diagnosis in terms of limited data. Also, by producing artificial data, they improve the quality and variety of educational sets and provide accurate information about the type, grade, and tumor behavior by extracting a minor characteristic of medical images, which helps diagnose early and better patients. In the following, some of the most important techniques in the field of BrT analysis are fully explained.

- **CNNs** CNNs specialize in image recognition tasks. They automatically learn spatial hierarchies in data, making them practical for detecting and classifying BrTs from MRI and CT images by identifying patterns like edges and textures. This technique is calculated using Eq. (1) [35].

$$y = \sigma \left(\sum_{i=1}^N W_i \times X_i + b \right) \quad (1)$$

In this equation, y is the outcome of the convolution operation, where W_i represents the convolutional filters (weights) for each image patch, X_i is the input image, and b is the bias term added to each layer. The $\sum$ denotes the summation of all the spatial locations and σ is the activation function, often the ReLU function, applied to the output.

- **Transfer Learning** Transfer Learning refers to using models already trained to solve new, related tasks with minimal data. This method is beneficial when dealing with limited data, often in medical imaging, where annotated datasets are scarce. This technique is calculated using Eq. (2) [36].

$$L = \frac{1}{M} \sum_{i=1}^M (y_{\text{true},i} - y_{\text{pred},i})^2 \quad (2)$$

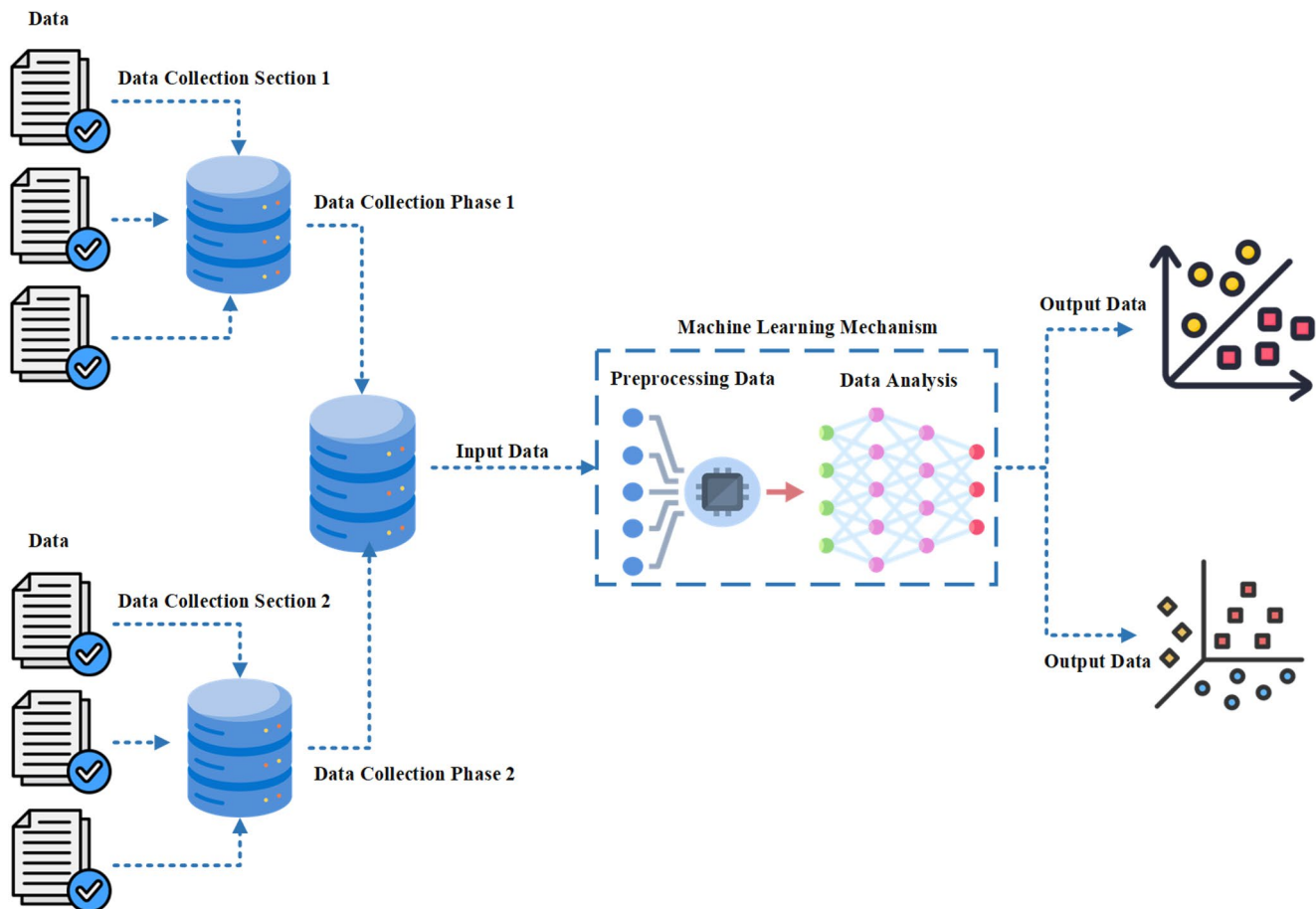


Fig. 4 ML mechanism

In this equation, L is the Mean Squared Error loss, where $y_{true,i}$ is the actual label and $y_{pred,i}$ is the predicted value. It calculates the disparity between predicted and actual values, directing model optimization.

- **Generative Adversarial Networks (GANs)** GANs are composed of two neural networks, a generator and a discriminator, that compete to improve data generation. For BrT detection, GANs generate synthetic medical images to augment training datasets, particularly when labeled data is scarce. This technique is calculated using Eq. (3) [37].

$$\min_Q \max_J V(J, Q) = \mathbb{E}_{x \sim p_{\text{data}}} [\log J(x)] + \mathbb{E}_{z \sim p_z} [\log(1 - J(Q(z)))] \quad (3)$$

In this equation, Q is the maker, and J is the discriminator. The loss function encourages J to distinguish real from generated data while Q improves to fool J into classifying fake data as real.

- **DL (Deep Neural Networks)** DNNs are multi-layered neural networks that excel in learning from large datasets. In BrT detection, they can identify intricate patterns in medical images, enabling precise classification of tumor types. This technique is calculated using Eq. (4) [38].

$$y = \sigma \left(W^{(l)} \cdot h^{(l-1)} + b^{(l)} \right) \quad (4)$$

In this equation, y represents the output of the current layer, $W^{(l)}$ denotes the weight matrix of layer l , $h^{(l-1)}$ is the output from the preceding layer, and $b^{(l)}$ is the bias term applied to the layer. The activation function σ , such as ReLU or Softmax, introduces non-linearity, which helps the model learn advanced patterns from the input.

- **Radiomics** Radiomics extracts quantitative features like texture and shape from medical images to help predict tumor characteristics and treatment response. These features provide valuable insights for BrT classification and prognosis. This technique is calculated using Eq. (5) [39].

$$\text{Feature} = \sum_{i=1}^N w_i x_i \quad (5)$$

In this equation, the Feature is the radiomic feature, W_i are the weights, and x_i are the extracted image

features. The summation aggregates the weighted contributions of each feature.

- **Support Vector Machines (SVMs)** SVMs are powerful for tumor classification. They find an optimal hyperplane that separates different tumor types in a high-dimensional space. They work well in scenarios with limited labeled medical data and provide robust generalization. This technique is calculated using Eq. (6) [40].

$$\min_{z,b} \frac{\|z\|^2}{2} \text{ s.t. } y_a(z \cdot r_a + b) \geq 1, \forall a \quad (6)$$

This equation minimizes the norm of the weight vector z , ensuring maximum margin separation between classes. The constraint ensures that each sample r_a with label y_a is correctly classified with a margin of at least 1.

- **FL** FL enables multiple decentralized medical institutions to train a model collaboratively without sharing raw patient data, enhancing privacy and security in BrT diagnosis. It aggregates local models into a global model while maintaining data confidentiality. This technique is calculated using Eq. (7) [41].

$$w_{t+1} = w_t - \eta \sum_{k=1}^K \frac{n_k}{N} \nabla L_k(w_t) \quad (7)$$

In this equation, w_{t+1} represents model parameters at iteration t , η is the learning rate, K is the number of clients (hospitals), n_k is the number of samples at client k , and $L_k(w_t)$ the local loss function is the global model updates by aggregating local gradients.

- **RL** RL optimizes tumor detection and treatment planning by learning policies that maximize long-term rewards. It is useful in adaptive radiotherapy and personalized treatment strategies. This technique is calculated using Eq. (8) [42].

$$Q(A_z, U_d) = Q(A_z, U_d) + n[r + \gamma \max_{za} Q(A_{za}, U_d n) - Q(A_z, U_d)] \quad (8)$$

In this equation, $Q(A_z, U_d)$ is the expected reward for taking action a in state s . α is the learning rate, r is the immediate reward, γ is the discount factor, and $\max_{za} Q(A_{za}, U_d n)$ estimates the best future reward. The model updates its knowledge iteratively to improve decision-making.

- **Supervised Learning (SL)** SL is widely used in medical imaging to classify tumors into benign or malignant based on labeled MRI scans. Models learn to map between input images and ground-truth labels. This technique is calculated using Eq. (9) [43].

$$L(W) = \frac{1}{N} \sum_{i=1}^N \mathcal{L}(f_W(x_i), y_i) \quad (9)$$

In this equation, $L(W)$ is the loss function minimized during training, $f_W(x_i)$ is the model's predicted output, y_i is the true label, and $\mathcal{L}$ represents a loss function like cross-entropy for classification.

- **Unsupervised Learning (USL)** USL is used for clustering tumor subtypes, anomaly detection, and feature extraction without requiring labeled data. It helps discover hidden patterns within MRI scans, segment tumor regions, and identify novel biomarkers for early diagnosis. This technique is calculated using Eq. (10) [44].

$$J = \sum_{i=1}^N \sum_{j=1}^K 1(c_i = j) \|x_i - \mu_j\|^2 \quad (10)$$

In this equation, the K-Means clustering objective function, where J minimizes intra-cluster distances. c_i assigns sample x_i to cluster j , and μ_j is the cluster centroid.

2.5 Validation Methods for ML Models in BrT Diagnosis

Evaluating the accuracy and generalizability of ML models is important in ensuring their performance in real-world situations. Without proper validation, models may perform well on test data but suffer performance degradation when exposed to new data. Therefore, various validation methods should be used to avoid overfitting and increase generalizability. In this study, three key methods are used to evaluate the performance of models:

- **Cross-Validation** Cross-validation is a statistical method for evaluating the performance of ML models that examines the model's generalizability to new data. This method divides the dataset into K folds, and the model is trained and tested multiple times on different parts of the data. One of the most common types is K-Fold Cross-Validation, in which the model is trained on $K-1$ folds of data and tested on the remaining fold. This process is repeated K times, and the average performance of the

model is calculated. This method reduces the dependence of the model on a specific dataset and prevents overfitting. In situations with limited data, the Leave-One-Out Cross-Validation (LOOCV) method is used, in which only one sample is used for testing in each iteration, and the rest of the data is used for training [45].

- **External Dataset Validation** External Dataset Validation refers to testing a model on a dataset not seen in the training process. This method checks the model's generalizability in real-world situations and avoids the model being dependent on specific data. External data is usually collected from independent sources, hospitals, or imaging devices to measure how well the model fits various data. This method helps identify problems related to data quality differences, patient population variations, and technical differences between imaging devices. If the model performs poorly on external data, it may indicate overfitting on the original training dataset [46].
- **Independent Testing on Unseen Data** In the independent testing method, the ML model is evaluated on data not seen during the training phase. This method examines the model's actual performance in clinical and real-world settings. It determines whether the model can generalize its learning to new data without relying on specific patterns in the training data. Independent testing is usually performed on a separate dataset from a hospital, imaging center, or other independent database. If the model performs poorly in this test, it may indicate over-reliance on specific training data or poor generalization. This method is important in medicine because any ML model must perform well on real data and new patients [47].

2.5.1 Interpretability of ML Models in BrT Diagnosis

In medical applications, the interpretability of ML models plays a crucial role in the acceptance and reliability of this technology in clinical settings. Understanding how models make decisions helps doctors better evaluate model predictions and leads to identifying possible biases and improving AI systems' performance. In this regard, several methods have been introduced for analyzing and interpreting ML models, and three key techniques in this area are reviewed below.

- **SHAP (Shapley Additive Explanations)** SHAP is a game-theoretic framework that evaluates the contribution of each feature to the model output. This method uses Shapley values to measure the influence of each feature on the final prediction and provides a comprehensive analysis of the model behavior. In BrT diagnosis, SHAP can be used to examine the impact of various

medical image features (such as MRI signal intensity in different brain regions) on the output of classification models. In addition to global interpretability, this method also provides the ability to examine the influence of features on individual predictions (Local Interpretability), allowing clinicians to understand the cause of specific predictions in each patient.

- **LIME (Local Interpretable Model-Agnostic Explanations)** LIME is a model-agnostic technique that helps to understand how the model makes decisions by making small changes to the input and analyzing the changes in the output. This method is very efficient for complex models that act as black boxes and can be used in deep learning models and SVM. In BrT diagnosis, LIME can be used to identify important parts of the image that have the most influence on the model's decision-making. This method forces the model to reveal its decision-making structure by generating artificial examples close to the original example. It helps increase the reliability of ML-based diagnoses by providing a simple and intuitive interpretation.
- **Grad-CAM (Gradient-Weighted Class Activation Mapping)** Grad-CAM is a gradient-based technique for interpreting deep learning models, especially CNNs, in medical image processing. This method produces gradient maps that show which areas of the image have the most influence on the model's prediction. In diagnosing BrTs using MRI images, Grad-CAM can highlight critical and damaged areas that play a key role in the model's decision-making process. This technique allows clinicians to see the relationship between image features and model output and discover irrelevant areas the model may have mistakenly used for prediction.
- Integrating these interpretability methods into ML models increases their transparency and reliability in diagnosing and treating BrTs. It allows them to be optimized based on clinical criteria.

2.5.2 Evolution Metrics

Evaluation of ML models in the diagnosis of BrTs includes a variety of criteria that examine the accuracy and reliability of the model. Accuracy (Accu) shows the correct predictions, but it is not sufficient in the set of unbalanced data. Precision (Pres) determines how many cases predicted as positive are accurate, while Recall (Reca) evaluates the amount of real positive samples identified. The F1 Score (F1s) is the average of precision and recall, providing better balance in uneven data. The Matthews Correlation Coefficient (MCC) considers all components of the intestinal matrix and provides a balanced evaluation of the model performance.

- **Accu** This evaluation metric is calculated as the ratio of correct predictions to the total number of samples. In imbalanced datasets, accuracy alone might not accurately represent model performance, as the model can be biased toward predicting the majority class. This metric is calculated using Eq. (11) [45].

$$\text{Accu} = \frac{\sum_{i=1}^N 1(y_i = \hat{y}_i)}{N} \quad \text{where } 1(y_i = \hat{y}_i) \in \{0, 1\} \quad (11)$$

This equation calculates accuracy as the fraction of accurate predictions from all the samples. The term $1(y_i = \hat{y}_i)$ is an indicator function that returns 1 when $y_i = \hat{y}_i$ and 0 otherwise.

- **Prec** Prec is a key metric for classification models, particularly when false positives are costly. It emphasizes the number of true positive predictions made by the model. This metric is especially critical in rare disease detection, where false positives could lead to unnecessary treatments or procedures. This metric is calculated using Eq. (12) [46].

$$\text{Prec} = \frac{\sum_{a=1}^N 1(Qz_a = 1 \wedge Q_a = 1)}{\sum_{a=1}^N 1(Qz_a = 1)} \quad \text{for } Qz_a \in \{0, 1\} \quad (12)$$

In this equation, precision is calculated as the ratio of true positives, where $(Q = Qz = 1)$ to the total number of positive predictions where $(Qz = 1)$. This metric is critical when the cost of a false positive is high.

- **Reca** Recall is a metric that indicates the percentage of actual positive samples correctly identified by the model. This metric is particularly important in medical diagnostics and rare disease detection, where identifying all positive cases is crucial to prevent missing critical diagnoses. This metric is calculated using Eq. (13) [47].

$$\text{Reca} = \frac{\sum_{i=1}^N 1(\hat{y}_i = 1 \wedge y_i = 1)}{\sum_{i=1}^N 1(y_i = 1)} \quad (13)$$

In this equation, y_i represents the predicted label for the i -th sample, and y_i is the true label. $1(y_i = 1 \wedge \hat{y}_i = 1)$ counts true positives, while $1(y_i = 1)$ counts all actual positive instances.

- **F1s** The F1s is the harmonic mean of precision and recall. It is beneficial when the data is imbalanced, and we must balance the importance of precision and recall. F1s measures the model's performance, especially when we want to balance false positives and negatives. This metric is calculated using Eq. (14) [48].

$$F_1s = 2 \times \frac{\text{Prec} \times \text{Reca}}{\text{Prec} + \text{Reca}} \quad (14)$$

In this equation, the F1s is calculated as the harmonic mean of Prec and ReCa. This metric is useful when we need to find a balance between Prec and ReCa, especially in imbalanced datasets.

- **MCC** The MCC is an advanced statistical metric that quantifies classification performance while addressing imbalanced datasets. This metric is calculated using Eq. (15) [49].

$$MCC = \frac{\sum_{i=1}^N [(TP_i \times TN_i - FP_i \times FN_i) \times \log(1 + TP_i + TN_i)]}{\sqrt{\prod_{j=1}^4 \left(\sum_{i=1}^N X_{ij} + \epsilon \right)^{w_j}}} \quad (15)$$

In this equation, MCC is computed by summing the weighted determinant of the confusion matrix, enhanced by a logarithmic correction to account for dataset size variations. The denominator incorporates a product of summations, ensuring robust normalization across different class distributions. Parameters w_j act as adaptive weighting factors, while ϵ prevents numerical instability.

3 Related Works

This section offers an in-depth analysis of prior research on using ML in cancer classification and BrT diagnosis. It aims to analyze the existing research, highlight key advances, and identify outstanding challenges in this field. By reviewing various methods, including DL techniques, feature selection methods, and classification models, it provides a structured view of the current state of the art. In addition, this section identifies research gaps and suggests potential avenues for future research to improve the accuracy and interpretability of ML models in medical applications.

Ali and Mohamad [50] showed that this investigation improved the feature selection method to increase the accuracy and significance of cancer subtype identification using a robust multinomial data clustering framework. They proposed a feature selection approach using cat swarm

optimization to extract relevant features for prediction. The results showed that after applying feature selection, the model's accuracy increased from 81 to 100%, demonstrating the superiority of this method over existing models and is a practical step towards understanding the molecular mechanisms associated with cancer subtypes.

Nazir et al. [51] conducted this study to offer an in-depth critical evaluation of recent research on BrT diagnosis and classification utilizing MRI scans and DL techniques. They reviewed past studies that utilized DL for this purpose. Then, they analyzed and compared the DL techniques proposed in articles published between 2015 and 2020 in a tabular format. Finally, they examined the advantages and disadvantages of deep neural networks and evaluated the effectiveness of these methods in diagnosing and classifying BrTs.

Mohamad and Ali [52] investigated the accuracy of cancer subtype classification by integrating multidimensional data and addressing the challenges of high dimensionality and limited sample size. They used Quantum Cat Swarm Optimization (QCSO) to select compelling features, applied the K-means algorithm for clustering, and employed the SVM for classification. Through QCSO, they identified important features, which led to an increase in model accuracy from 81 to 100%. They evaluated the model performance using criteria such as accuracy, F1-score, sensitivity, specificity, ROC, and silhouette index. The results confirmed the effectiveness of their feature selection method.

Amin et al. [53] showed that this study provides a comprehensive review of research related to BrT diagnosis using MRI to assist researchers in this field. This study reviewed various aspects, including BrT anatomy, public datasets, image enhancement techniques, segmentation, feature extraction, classification, DL, transfer learning, and quantum ML for BrT analysis. Finally, this study provided a comprehensive reference for researchers by examining the advantages, limitations, recent advances, and future trends in BrT diagnosis.

Ali and Mohamad [54] reviewed the latest studies on using AI in omics data analysis and identified the fundamental challenges in this field. To achieve this goal, they used a systematic strategy to search for articles related to AI and omics using different combinations of keywords. The findings of this study provided a comprehensive view of the intersection of the omics fields and AI models. They could be valuable for professionals seeking a general understanding of the application of AI in omics data processing.

Soomro et al. [55] conducted this study to offer a detailed overview of research related to BrT segmentation in MRI images, aiming to aid in the rapid and accurate diagnosis of this disease. They reviewed articles published between 1998 and 2020 and analyzed the main segmentation algorithms.

They used machine image segmentation and DL to identify BrTs. Their findings demonstrated that DL methods outperformed other techniques in tumor segmentation from MRI images.

Habibi et al. [56] conducted this study to investigate and evaluate the performance of ML algorithms in accurately distinguishing glioblastoma from metastatic BrTs. They systematically reviewed published studies up to June 2023 and analyzed data from 28 studies with 2860 patients. They found that DL algorithms, including ResNet50 and ResNet50-LSTM, accurately distinguished these two tumor types. The study showed that these methods achieved a sensitivity of 83% and spec of 87%, with an area under the curve (AUC) of 0.90, indicating their high ability to discriminate between the tumors.

Table 1 outlines and compares review papers about implementing ML in diagnosing cancer and BrTs. It includes each study's goals, advantages, challenges, years, publications, and applications. The goal is to offer a thorough perspective on the progress in this field, identify existing limitations,

and help researchers better understand research trends and future directions.

4 Methodology of Research

This section illustrates the criteria for selecting research articles using ML techniques to identify and cure BrTs. It gives researchers a structured approach to acquiring, organizing, and investigating research questions. It also highlights key points in the field and examines the results of studies in detail.

4.1 Crafting Research Inquiries

This study investigates the objectives, benefits, limitations, innovations, evaluation criteria, and ML methods used in identifying and curing BrTs in published articles over a specified period. It also analyzes the applications of these methods in improving diagnostic and therapeutic processes. This investigation seeks to evaluate the effectiveness of

Table 1 Summary of review papers on ML applications in cancer and BrT detection

References	Goals	Advantages	Challenges (Disadvantages)	Applications	Publisher	Year
[50]	Improving feature selection for identifying cancer subtypes using multi-omics data clustering	Increased classification accuracy to 100%, improved model interpretability,	Limited availability of high-quality datasets, difficulty in comparing different feature selection methods	Biological research, precise classification of cancer subtypes, understanding molecular mechanisms of cancer	JSCDM	2024
[51]	Critical analysis of DL methods for BrT detection and classification using MRI images	A comprehensive comparison of DL methods assists researchers in selecting the best approach.	Lack of uniform evaluation metrics across studies, limited generalizability due to dataset differences	DL research, improving BrT detection from MRI	Elsevier	2021
[52]	Using Quantum Cat Swarm Optimization for feature selection and improving cancer classification accuracy	Accuracy increases from 81–100%, enhanced model interpretability, and better insights into biological mechanisms.	The challenge of integrating findings from different studies, inconsistencies in reported accuracy metrics	Biological research, improving cancer diagnosis accuracy, applications in biological modeling	IJCSM	2024
[53]	A comprehensive survey on BrT detection through MRI and analysis of ML techniques	Extensive coverage of different techniques, review of future trends, a valuable resource for researchers	Lack of a standardized benchmark for evaluating methods, difficulty in synthesizing conclusions across studies	Medical research, BrT detection, analysis of ML-based methods	Springer	2022
[54]	Evaluating AI methods in omics data analysis and identifying key challenges	Provides a broad perspective on AI applications in omics, highlights key challenges	Variability in AI model performance across different studies, lack of real-world validation of proposed solutions	Omics data analysis, AI applications in biology	IJMCS	2024
[55]	Reviewing various MRI image segmentation techniques for BrT detection	Comprehensive analysis of different segmentation algorithms, identifying practical approaches	Differences in segmentation evaluation criteria, lack of a unified framework for comparing results	Early detection of BrTs, improving medical image segmentation methods	IEEE	2022
[56]	Analyzing ML algorithms for differentiating glioblastoma from metastatic BrTs	High diagnostic accuracy (sensitivity: 83%, specificity: 87%), improved performance using CNN-RNN integration	Limited reproducibility of ML model results across different datasets, challenges in standardizing evaluation criteria.	Differential diagnosis of BrTs, improving accuracy in medical image evaluation	Springer	2025

these techniques and identify the associated challenges in managing this disease. In addition, the research primary questions (PQs) designed to accomplish these goals are provided in Table 2 to understand this complex area better.

4.2 Selecting and Evaluating Articles

This study adopts a systematic and rigorous approach to reviewing, selecting, and evaluating scientific articles to offer a comprehensive picture of the use of ML techniques in the therapy and detection of BrTs. In February 2025, a systematic search was conducted in reputable scientific databases to ensure maximum coverage of reliable sources. The search included articles published between 2023 to February 2025 that attempted to use a diverse set of keywords to find valuable and evidence-based research. This keyword consists of (“Machine Learning” OR “Deep Learning,” “Supervised Learning” OR, “Reinforcement Learning” OR “Unsupervised Learning,” OR “Brain Tumor Detection,” OR “Brain Tumor Segmentation” OR “Glioblastoma” OR “Meningioma” OR “Pituitary Tumor” OR “Magnetic Resonance Imaging (MRI)” OR “Computed Tomography (CT)” OR “Brain Tumor Classification”) Also, the search included the following combinations: (“Machine Learning in Brain Tumor Diagnosis”) OR (“Deep Learning in Brain Tumor Analysis”) OR (“Supervised Learning in Brain Tumor

Detection”) OR (“Unsupervised Learning in Brain Tumor Segmentation”) OR (“Neural Networks for Brain Tumor”) OR (“Brain Tumor MRI Processing”) OR (“AI-Based Brain Tumor Classification”). Selecting articles for a review on ML applications in identifying and curing BrTs began with collecting articles from various scientific databases (such as Elsevier, Springer, IEEE, etc.). We reviewed only articles published between 2023 and 2025. Out of the initial 550 articles, we first evaluated their relevance to ML in BrTs and removed 320 irrelevant articles. Then, we assessed 230 relevant articles for quality and removed 130 articles, including 114 inappropriate articles, four review articles, two books, and 10 duplicate articles. Subsequently, we retained 100 high-quality and relevant articles, reviewed them for their relevance to ML, and removed 28 irrelevant articles. After this stage, we also evaluated the articles’ relevance to BrTs and removed another 37. Finally, the researchers selected 35 high-quality articles related to ML and BrTs as the final articles for the review article. Figure 5 illustrates the detailed process of selecting and excluding articles.

Table 3 presents the most relevant papers on the use of ML in identifying and curing BrTs, which have been investigated in highly regarded academic publications such as Elsevier, Springer, IEEE Explorer, Nature, MDPI, Wiley, and Other – The journals categorized under “Other” include those that are not as well-known or prominent as the other listed journals. Therefore, they have been grouped under the “Other” category for simplicity.

Figure 6 accurately displays the distribution of the 35 chosen studies from 2023 to February 2025. The graph shows that the most significant number of articles were published in 2024 and 2025, indicating a substantial increase in the interest of researchers in this domain in the past few years. This growing trend in the publication of studies suggests the expansion of research and advances in employing ML techniques to detect and therapy BrTs. It emphasizes the increasing importance of this field in the scientific community. In addition, this figure shows the spread of articles released by various publishers during this time frame. The results show that IEEE has the largest share of articles. Therefore, this indicates the key role of IEEE in publishing scientific and research articles related to this field and shows that this publisher has contributed more than others to advancing studies.

Table 2 Crafting research inquiries

PQs	Question	Explanation
PQ1	What are the primary objectives of using ML in BrT therapy and detection?	The aim is to explore the key goals behind applying ML to improve diagnostic and treatment processes. Section six will explore these objectives in detail.
PQ2	What are the main advantages of ML compared to traditional methods in BrT therapy and detection?	The focus is on analyzing ML’s benefits, particularly its improvement in accuracy, efficiency, and adaptability. Section six will provide a comprehensive review.
PQ3	What are the major drawbacks of using ML techniques BrT therapy and detection?	This inquiry seeks to identify obstacles and issues related to applying these methods to BrT research. Section six will discuss these drawbacks.
PQ4	What are the applications of ML for BrT therapy and detection?	The purpose is to explore recent applications shaping the field. Section six will highlight these applications.
PQ5	What are the key evaluation metrics for assessing ML models in BrT therapy and detection?	The goal is to review essential criteria for evaluating the accuracy and effectiveness of ML models. Section six will examine these metrics.
PQ6	What strategies can enhance the performance of ML models in BrT therapy and detection?	The intention is to propose methods for optimizing models, addressing challenges, and improving efficiency. Section seven will discuss these strategies.

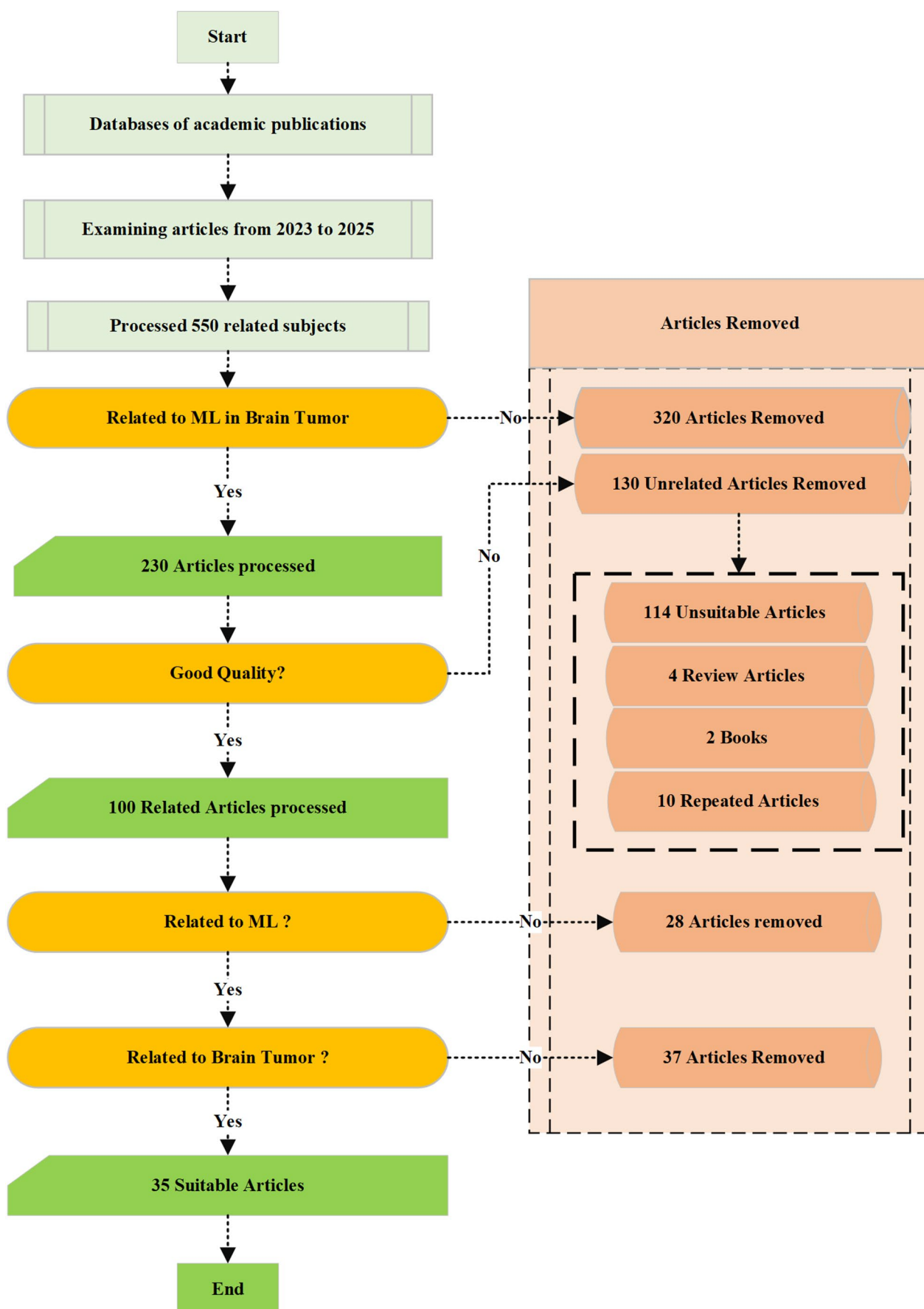


Fig. 5 Selection and review process of high-quality articles on ML applications in identifying and curing BrTs

5 A Systematic Taxonomy of ML Applications in BrT Identification and Treatment

ML is evolving with widespread applications in detecting and treating BrTs, primarily by analyzing medical images. Hence, this technology can use advanced algorithms to simulate hidden patterns and complex relationships between tumor features and achieve more accurate analyses [57]. Thus, this increases the accuracy of detecting and diagnosing tumors, especially in the early stages and in low-resolution medical images. In addition, ML significantly impacts the treatment process of patients with BrTs by optimizing clinical decisions and suggesting personalized treatments [58]. These techniques also allow doctors to better predict the course of therapy and dynamically adjust treatment strategies. Therefore, ML, as an advanced tool, improves the process of diagnosing, classifying, and treating BrTs, helping to improve the accuracy and efficiency of medical care [59]. In this review paper, which investigates the use of ML techniques in identifying and curing BrTs, the authors present a systematic taxonomy consisting of four main sections. This taxonomy includes FL, RL, DL, and hybrid techniques. Hybrid learning refers to a set of techniques that do not fall into the categories of FL, RL, and DL, and it includes methods that incorporate the characteristics of different techniques, including SL and USL, that researchers have used. The authors display this taxonomy in Fig. 7. The authors adopt this taxonomy to accurately and comprehensively categorize different ML techniques for identifying and curing BrTs so that researchers can quickly identify and analyze the appropriate tools and approaches for each type of application. This taxonomy aims to facilitate the selection and application of different techniques to improve the accuracy of diagnosis, personalize treatments, and increase the efficiency of clinical processes in this field.

5.1 Exploring FL Techniques for BrT Identification and Treatment

The power of FL techniques in identifying and treating BrTs is a novel solution in medical AI, allowing the processing of medical imaging data without direct transfer to a central server. This approach preserves patients' privacy and enables the training of DL models on data distributed across different hospitals. In this regard, advanced techniques such as gradient FL optimization, local model aggregation using FedAvg, and FLProx customization are employed to enhance the accuracy of BrT diagnosis. Combining FL with CNNs and transformer-based models increases accuracy

and reduces the error rate in BrT classification. Additionally, integrating security enhancement techniques such as homomorphic encryption and secure multi-party computation (SMPC) into FL enables the secure exchange of gradients, thereby improving the reliability of diagnostic models. However, challenges such as synchronizing heterogeneous models, handling data distribution heterogeneity, and managing computational overhead remain significant obstacles to implementing FL for BrT identification and treatment. Hence, this underscores the need for further research and the development of scalable optimization methods, which will be crucial for the future of FL in medical AI. This section will discuss articles that have tried to use FL techniques to identify and cure BrTs.

Appasami and Savarimuthu [60] investigated improving brain MRI image recognition and classification accuracy and efficiency using transfer and FL. In this method, transfer learning utilized the features of pre-trained models to increase the convergence speed and improve performance. The proposed method, which combined FL and VGG-based deep architecture, achieved an accuracy of 98.4% in analyzing medical MRI images. These results showed that this approach provided high accuracy in medical image classification and met the data security requirements.

Maiti et al. [61] investigated the development of an advanced model based on the ResNet-10 convolutional neural network to improve the classification of brain CT scan images and diagnose diseases such as Alzheimer's, brain lesions, and normal aging in innovative health. This study used associative learning and interpretive generative AI (GAI) to increase the efficiency in diagnosing BrTs. These findings indicated that the suggested model not only achieved well in the diagnosis of BrTs but also had the potential to be used in collaborative learning based on generative AI.

Wu et al. [62] investigated the development of an advanced FL-based model to improve the classification of BrT MRI images and solve the problems caused by heterogeneous data distribution. This study introduced a new framework called FedSPD, which improved the coordination between local models and global knowledge by utilizing similarity-preserving knowledge distillation. Extensive experiments on several datasets showed that FedSPD improved accuracy by 78.41% compared to traditional FL methods and increased computational efficiency by reducing the parameter size and training time.

Albalawi et al. [63] investigated the development of a DL model based on FL to improve the classification of BrTs in MRI images. Experiments conducted on figshare, SARTAJ, and Br35H datasets indicated that the suggested model had high accuracy in classifying types of BrTs and performed better than existing methods. Finally, this model effectively

Table 3 Trustworthy academic databases

Database	Database References	Publication Count (2023–2025)
Elsevier	https://www.elsevier.com	4
Springer	https://www.springer.com	8
IEEE Explorer	http://ieeexplore.ieee.org	9
Nature	https://www.nature.com	2
MDPI	https://www.mdpi.com	2
Wiley	https://www.wiley.com	5
Other	–	6

improved the automatic diagnosis of BrTs and increased the accuracy of clinical decisions by achieving 98% accuracy.

Kumar et al. [64] developed a framework based on FL and permissioned blockchain to improve BrT segmentation accuracy and maintain medical data privacy. This method allowed medical centers to train a central model without sharing raw data and improved the model's accuracy in diagnosing different diseases. Experiments showed that the proposed algorithm increased the Dice similarity coefficient by 1.99% and reduced the Hausdorff distance for complete tumors by 19.08% compared to the baseline methods.

Zhang et al. [65] investigated a novel approach based on FL to improve BrT diagnosis and preserve patient data privacy. In this method, each medical center, as a participant, maintained its local data and sent part of its model parameters to a central server for knowledge aggregation and sharing through secure communication. Experimental results showed that this method performed better than centralized learning while accurately preserving data privacy.

Onaizah et al. [66] investigated a DL-based approach to improve BrT detection and address the challenges of protecting patient data privacy. For this purpose, a Siamese convolutional neural network (SiCNN) was designed and

implemented in a federated peer-to-peer (P2P) learning environment to enable collaboration between local systems without direct data sharing. The evaluation showed that the SiCNN model performed well in BrT detection and provided high accuracy while ensuring data safety in the DL process.

Yahiaoui et al. [67] investigated FL with privacy-preserving techniques for BrT lesion segmentation using a 3D U-Net model in a distributed environment. This study used differential privacy techniques to address privacy and data-sharing challenges. The findings demonstrated that the proposed model operated very well in BrT segmentation in the final experiment, achieving DSCs of 89.85%, 87.55%, and 86.6% while preserving the privacy of patients' data.

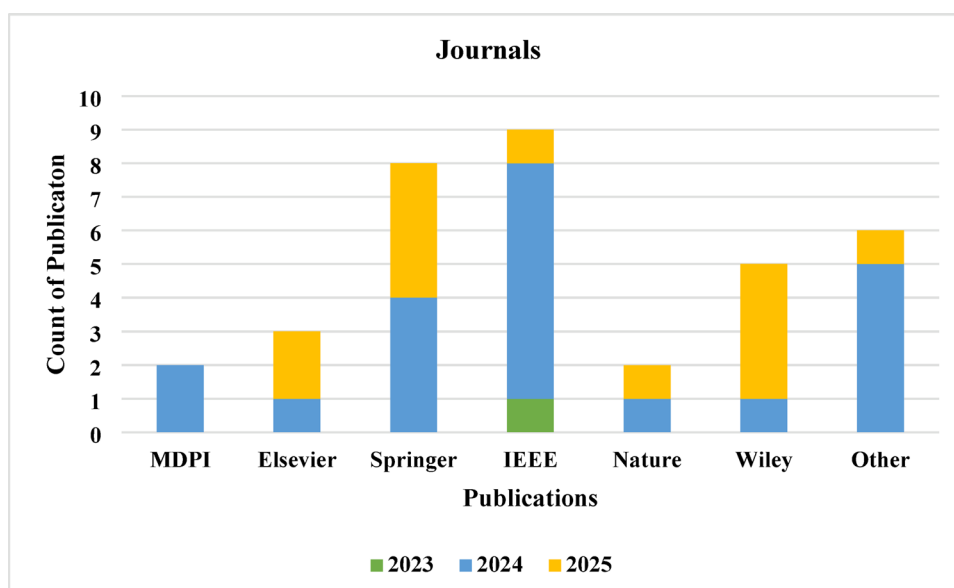
Ravali et al. [68] investigated the accurate and automated classification of BrTs using MRI images to overcome the challenges of tumor shape complexity and imaging variations that hinder precise diagnosis. This model used FL and transfer learning methods, and this model, the VGG16 network, was tuned explicitly for brain MRI images. This model architecture improved its ability to classify BrTs using pre-trained neural networks. Finally, the proposed model successfully diagnosed different types of tumors with an accuracy of 98.5%.

Table 4 aims to provide an overall comparison of different studies that have used FL techniques in the therapy and detection of BrTs. By highlighting the strengths and weaknesses of each approach, it offers an in-depth picture of the trends and challenges in this field.

5.1.1 Findings and Discussions

The data analysis in Table 4 reveals that researchers have applied FL to identify and cure BrTs. They have used FL

Fig. 6 Publication trends on ML applications in BrT therapy and detection (2023–2025)



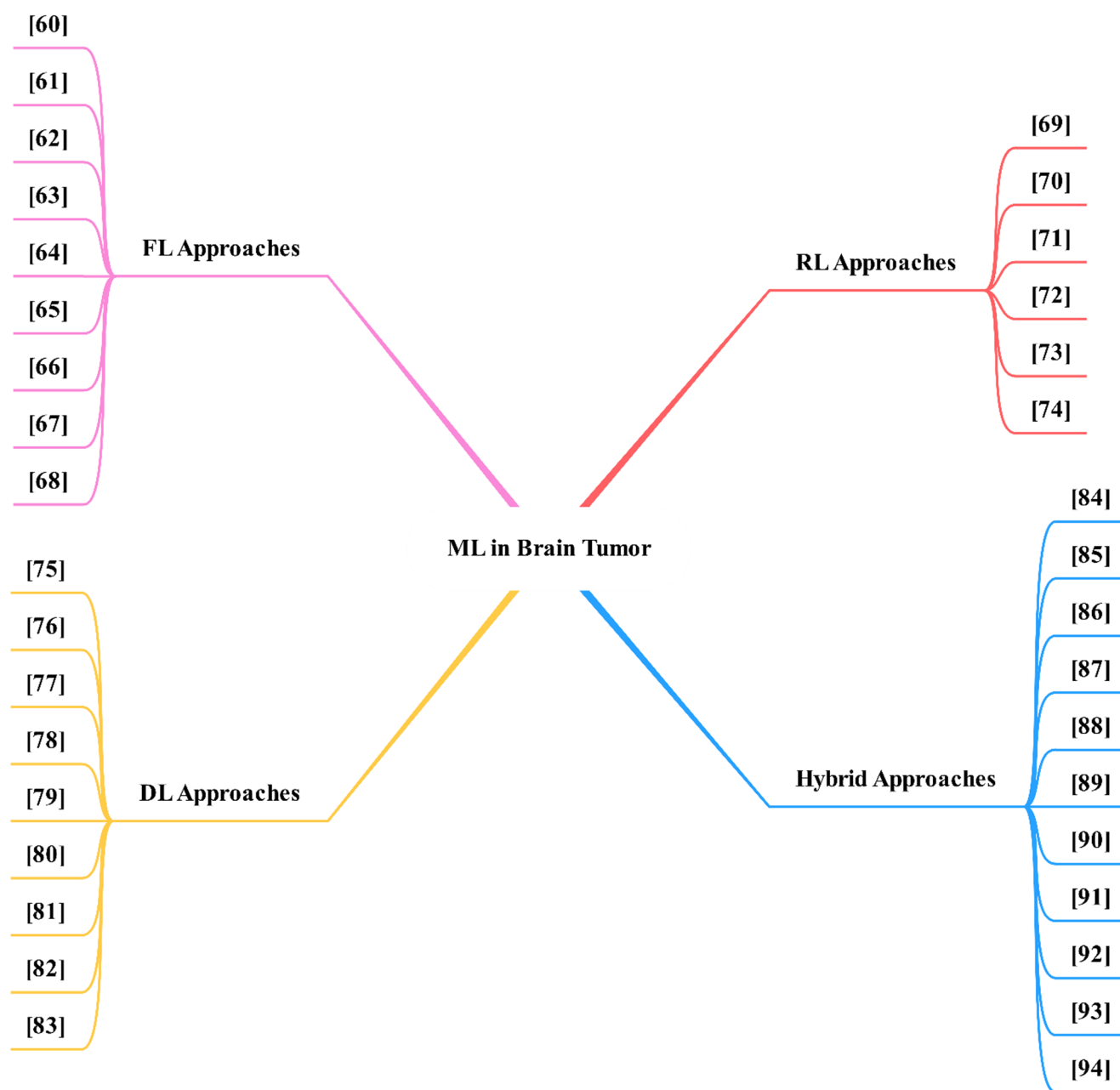


Fig. 7 A systematic taxonomy of ML applications in BrT identification and treatment

for various objectives, including classifying MRI and CT images, improving accuracy in non-IID conditions, preserving privacy, accelerating model convergence, and increasing processing efficiency. The methods used include Transfer Learning, FedSPD, Peer-to-Peer FL, Federated Migration Learning, and using blockchain to increase data security. The main advantages of FL include improving accuracy, preserving privacy, reducing processing costs, and improving performance in decentralized environments. In contrast, the disadvantages include security challenges, architectural complexity, the need for fine-tuning for heterogeneous data,

and increased training time in some cases. Regarding evaluation criteria, studies have mainly emphasized Accu, Prec, Reca, F1s, and MCC. In some cases, the Dice Coefficient and Hausdorff Distance have also been used to evaluate segmentation performance. Studies show that FL is a practical approach for processing decentralized medical data. However, it still needs optimization in areas such as reducing computational complexity and improving model adaptation to heterogeneous data.

Table 4 Comparison of FL techniques for BrT therapy and detection

References	Objectives	Advantages	Drawbacks	FL techniques	Application	Evaluation metrics
[60]	Use transfer learning within FL to improve MRI brain image classification.	Faster convergence, privacy preservation, and high performance.	Concerns regarding the security of sensitive data.	Federated Transfer Learning; CNN based on VGG.	BrT diagnosis in MRI images.	Accu
[61]	Classify brain CT images to diagnose tumors (and other brain disorders).	Utilizes ResNet-10 with RHAM for precise feature extraction and high accuracy.	It does not employ an FL approach; limited training samples may affect performance.	ResNet-10 (Focuses on CNN and GAI techniques)	BrT and other brain disorder diagnosis.	Accu
[62]	Address Non-IID data challenges in FL for BrT MRI images.	Significant accuracy improvement over traditional FL and PFL; reduced computational cost (up to 67.25% per training round).	Limitations of PFL, such as lower accuracy or high memory consumption, require fine-tuning for heterogeneous data.	FedSPD: Personalized FL with Similarity-Preserving Knowledge Distillation.	BrT classification in MRI images.	Accu: Reported improvement (up to ~77.77–78.41% over baselines);
[63]	Automate BrT classification using FL and transfer learning.	Very high accuracy; improved precision and F1-score; minimal manual intervention.	Lacks discussion on FL communication challenges; requires careful model tuning for complex data.	FL combined with transfer learning; modified VGG16 architecture.	Classification of various BrT types (glioma, meningioma, pituitary, etc.).	Accu, Prec;
[64]	Improve BrT segmentation while preserving privacy.	Enhanced segmentation performance with improved Dice and reduced Hausdorff distance; uses blockchain for parameter encryption.	Complex architecture: asynchronous FL may increase training time.	Blockchain-based FL (permissioned blockchain); parameter masking.	BrT segmentation in MRI images.	Dice coefficient (+1.99% improvement); Hausdorff distance (–19.08% reduction); Accu, prec, Accu
[65]	Detect BrTs using FL for secure multicenter data sharing.	Maintains data privacy; improved diagnostic performance compared to centralized approaches;	Limited details on specific drawbacks; potential performance drop with highly variable data.	Federated Migration Learning; partial parameter sharing.	BrT detection in multicenter environments.	Accu
[66]	Provide a SiCNN-based approach in a P2P FL environment for BrT diagnosis.	Ensures privacy and acceptable performance in tumor detection using SiCNN.	Lacks precise numerical results and comprehensive performance comparisons.	Peer-to-Peer FL; SiCNN model.	BrT classification from MRI images.	Accu, Prec, F1s
[67]	Precisely segment BrT lesions while preserving privacy.	Good performance in terms of DSC and Hausdorff; utilizes privacy-preserving techniques for enhanced security.	Limited analysis on PPT drawbacks; potential computational overhead due to privacy techniques.	FL with 3D U-Net; Differential Privacy.	Segmentation of tumor lesions (WT, TC, ET) in MRI images.	DSC (89.85%, 87.55%, 86.6% for WT, TC, ET respectively); Hausdorff distance (22.95 mm, 8.68 mm, 8.32 mm); Accu, prec, MCC
[68]	Automatically classify BrTs using FL and transfer learning.	Very high accuracy; leverages pre-trained models for improved performance.	Lacks comprehensive discussion on FL challenges with non-IID data; further evaluation in real-world scenarios is needed.	FL with transfer learning; modified VGG16.	BrT diagnosis and classification in MRI images.	Accu, F1s

5.2 Exploring RL Techniques for BrT Identification and Treatment

RL techniques in identifying and treating BrTs are one of the most advanced medical AI approaches, optimizing decision-making through continuous environmental interaction and improving diagnostic and therapeutic strategies. These techniques, using Proximal Policy Optimization (PPO), Deep Q-Networks (DQN), and ACTOR-Critic Methods models, can direct key processes such as determining tumor boundaries in MRI images, off-the-invasive surgical routes, dose adjustment, and optimization of dietary diets. On the other hand, RL integration with DL methods such as CNNs and transformer-based models improves the accuracy of classification and prediction and reduces the error rate. One of the key applications of RL in this area is to personalize therapeutic methods based on individual responses, which is done by modeling dynamic rewards and regulating adaptive learning policies. In addition, the combination of RL with FL methods enables the processing of distributed data among various medical centers without violating patient privacy. However, challenges such as instability in the learning process, the need for a high volume of label data, computational complexity, and high costs of training RL models remain obstacles to the widespread implementation of these techniques in identifying and curing BrTs. Developing scalable optimization methods, improving convergence strategies, and using distributed models to reduce computational overhead are key strategies to overcome these challenges and improve RL's efficiency in smart medicine. This section will discuss articles that have tried to use RL techniques to identify and cure BrTs.

Shekhar et al. [69] investigated improving the accuracy and efficiency of DL models in BrT detection from MRI images by using an innovative adaptive learning rate scheduler. For this purpose, RL concepts were used to adjust the learning rate during the neural network's training process. This method continuously interacted with the RL agent to dynamically update the learning rate. The results showed that this method achieved better performance in terms of accuracy and efficiency than traditional learning rate techniques.

Mani et al. [70] investigated the rapid and accurate detection of BrTs using image processing and RL techniques. This method divided each MRI image into 8×8 blocks and extracted features such as mean, variance, skewness, kurtosis, entropy, contrast, correlation, energy, and homogeneity. The results showed that the proposed algorithm's accuracy was 97%, its sensitivity was 96%, and its specificity was 97%, indicating the high efficiency of this method in classifying BrT MRI images.

Sharvani et al. [71] investigated improving anatomical reference point recognition in medical images using multi-agent RL. For this purpose, a Deep Q-Network architecture with connected agents that cooperated during the training process was used. The framework supports multiple agents, each responsible for identifying specific reference points and improving the recognition accuracy. The results showed that this method reduced the recognition error by 50%, requiring less computational resources and training time.

Costanzo and Flores [72] investigated how RL could improve tumor detection in microwave imaging. They proposed a new framework to combine RL with microwave imaging techniques to enhance tumor reconstruction. They performed numerical validation using BrT images to demonstrate the effectiveness of the proposed framework. The findings revealed that their method delivered high accuracy in detecting single and multiple tumors, with slightly better accuracy in evaluating the criteria for the former case.

Chaki and Woźniak [73] investigated the improvement of BrT image classification and retrieval using MRI images and RL methods. For this purpose, they introduced an RL agent that can interact with an environment containing BrT images and retrieve and classify images similar to an unknown requested image. The findings demonstrated that the proposed method exhibited excellent accuracy in BrT image classification and retrieval, with 97.1% for meningioma, 98.7% for glioma, 94.3% for pituitary tumors, and 100% for non-tumor images, respectively.

Khan et al. [74] improved peer selection in dynamic and distributed FL environments using RL methods. In this study, they introduced a new algorithm called RL-HSimAgg that uses RL and similarity-weighted aggregation methods to handle outliers. They conducted simulation experiments on internal and external datasets and found that the RL-HSimAgg algorithm achieved better results than the EG method using UCB.

Table 5 aims to provide an overall comparison of studies that have used RL techniques in the therapy and detection of BrTs. By highlighting the strengths and weaknesses of each approach, it offers an in-depth picture of the trends and challenges in this field.

5.2.1 Findings and Discussions

Data analysis Table 5 shows that RL techniques have helped improve accuracy, reduce errors, and increase models' efficiency in BrT therapy and detection. Some studies have provided faster and more stable convergence in CNN for BrT diagnosis using an RL-based adaptive learning rate scheduler. Other methods have used Q-learning to extract accurate features and improve classification accuracy, but they have limitations due to the complex process of feature extraction

Table 5 Comparison of RL techniques for BrT therapy and detection

References	Objectives	Advantages	Drawbacks	RL techniques	Application	Evaluation metrics
[69]	Propose an RL-based adaptive learning rate scheduler for CNN training in BrT detection from MRI scans.	Dynamically adjusts learning rate, faster and more stable convergence, improved accuracy and efficiency.	No specific numerical results were provided, only relative performance improvement.	RL-based adaptive learning rate scheduler integrated into CNN.	BrT detection from MRI.	Accu, F1s
[70]	Utilize Q-learning to classify BrT MRI images using block-based feature extraction.	Detailed feature extraction; competitive performance with high accuracy.	Complex feature extraction process and block division; limited to static feature descriptors.	Q-learning with a reward mechanism.	BrT classification from MRI.	Accu, Reca, F1s: (Additional: Specificity: 97%)
[71]	Apply multi-agent RL for anatomical landmark detection in brain imaging.	Reduces detection errors by 50%; requires fewer computational resources and less training time.	Focuses on landmark detection rather than direct tumor diagnosis; standard evaluation metrics are not provided.	Deep Q-Network with collaborative multi-agent framework.	Anatomical landmark identification in brain imaging.	Accu, Reca, F1s, (Error reduction: 50%)
[72]	Integrate RL with microwave imaging to optimize tumor reconstruction parameters.	Enhances adaptive tumor identification; improves reconstruction performance.	Evaluated on breast images (conceptually transferable); lacks direct standard metric values for BrTs.	Deep Q-Network (DQN) is combined with the Quadratic Born Iterative Method and U-Net.	Tumor reconstruction (applicable to BrTs conceptually).	Accu; Prec; (Additional: Comparison between single and multi-tumor cases)
[73]	Develop an RL agent for BrT categorization and retrieval using a fuzzy logic-enhanced CNN.	Achieves high accuracy across different tumor types, innovative fuzzy logic, and RL fusion.	Complex architecture with extensive preprocessing; binary coding via MPCA adds extra steps.	DBIRA2.0-RLN (CNN-based RL agent) with fuzzy logic enhancement.	BrT categorization and retrieval in MRI.	Accuracy: Meningioma 97.1%, Glioma 98.7%, Pituitary 94.3%, No tumor 100%; Reca
[74]	Use RL for collaborator selection in FL to improve brain lesion segmentation.	Enhances model robustness; UCB method outperforms epsilon-greedy; efficient client selection.	It focuses on the FL environment rather than pure RL, and the evaluation is based on segmentation (Dice) rather than standard classification metrics.	Multi-armed bandit algorithms (Epsilon-Greedy and UCB) for collaborator management.	Federated brain lesion segmentation in multi-modal MRIs.	Accu, Prec, (Additional: Dice scores – Enhancing Tumor: 0.7334, Tumor Core: 0.7432, Whole Tumor: 0.8252)

and image block segmentation. Some models have also used multi-agent RL to identify anatomical landmarks, which reduces the detection error, but their focus is not on direct tumor detection. In some studies, Deep Q-Network (DQN) is combined with microwave imaging to provide more accurate tumor reconstruction. However, researchers have primarily tested this method on breast cancer images, and they need to investigate its applicability to BrTs further. Some studies have explored the combination of fuzzy logic and CNN-based RL for BrT classification. This approach accurately identifies tumor types but involves a complex architecture and requires extensive data processing. Also, some methods have used multi-armed bandit algorithms to optimize participant selection in FL, which improves model accuracy; their focus is more on segmentation than classification. Overall, using RL in identifying and curing BrTs has brought several advantages, such as improved accuracy and reduced processing time. However, there are challenges,

such as computational complexity, the need for high-quality data, and limitations in some evaluation criteria.

5.3 Exploring DL Techniques for BrT Identification and Treatment

In BrT therapy and detection, using DL techniques as an advanced approach to processing medical images has improved the precision and efficiency of diagnostic systems. CNNs and Transformer-based models play a key role in accurately identifying BrTs due to their ability to extract complex features from MRI images. Also, Transfer Learning models provide an optimal classification of tumors using pre-taught architectures such as Resnet and EfficientNet. In this regard, the DL combination with data augmentation and precise fine-tuning reduces the error rate and increases the generalization of diagnostic models. In addition, integrating optimization techniques such as ADAM and Lamb in the

learning process has increased the convergence of models and improved speed and accuracy. However, challenges such as data imbalances, the need for heavy calculations, and the explanation of DL models are still fundamental obstacles to the widespread implementation of these techniques in clinical systems that require further research and development of DL model optimization methods. This section will discuss articles that have tried to use DL techniques to identify and cure BrTs.

Mansur et al. [75] investigated different methods for BrT segmentation and tumor region identification in MRI images. This study examined three segmentation methods: threshold-driven segmentation, region-focused segmentation, and using CNNs with Unet architecture for BrT segmentation. The results showed that the Dice similarity parameter for threshold-based partitioning was 0.99, for region-based segmentation, it was 0.878, and for segmentation using Unet, it was 0.57.

Ali et al. [76] investigated using DL techniques to improve accuracy in BrT classification. For this purpose, the pre-trained EfficientNetB3 and DenseNet121 models were used in combination with the coevolutionary genetic algorithm (CEGA) to optimize hyperparameters and classify four types of BrTs, including gliomas, meningiomas, pituitary adenomas, and no cancer. The experimental results showed that the CEGA-EfficientNetB3 model achieved 99.39% accuracy, and the CEGA-DenseNet121 model achieved 99.01% without data augmentation.

Mugdha and Uddin [77] investigated the comparison of a new convolutional neural network (CNN) model with pre-trained models via transfer learning. This study comprehensively compared VGG-16, ResNet-50, AlexNet, and Inception-v3 models. The results showed that the VGG-16 model performed best, with 95.52% accuracy in testing, 99.87% in training, and a confidence loss of 0.2348. The ResNet-50 model was next, with 93.31% accuracy in testing and a confidence loss of 0.6327.

Elhadidy et al. [78] investigated a comprehensive method for BrT classification using MRI images and DL models. They classified MRI images into four categories: healthy brains, and they used normalization, resizing, and data augmentation preprocessing. They then trained each DL model on the preprocessed data and evaluated their performance using precision, sensitivity, and specificity. The results showed that the SwinTransformer and EfficientNet models performed better, achieving experimental accuracy of 98.08% and 98.72%, respectively, compared to the conventional CNNs, which achieved an accuracy of 95.16%.

AlShowarah [79] investigated the potential for identifying brain cancer from images using deep learning methods and statistical techniques. In this study, the features were extracted using VGG-19 and AlexNet CNN models and

then used to create fresh statistical procedure datasets. Finally, the results showed that statistical operations, especially for large values, and feature fusion using KNN 99.1% and SVM 99.1% gave the best accuracy.

Gencer and Gencer [80] investigated DL-based methods for automatic multiclass classification of BrTs. They proposed a novel approach combining DL and quantum genetic algorithms (QGA). In this approach, they utilized the pre-trained feature extraction power of the EfficientNetB0 model and improved feature selection with the help of quantum genetic algorithms. The results showed that this hybrid model achieved high accuracies of 98.36% and 98.25% with different datasets and significantly outperformed traditional methods.

Hosny et al. [81] investigated DL methods for BrT detection and classification. They proposed a hybrid approach using transfer learning and pre-trained DenseNet121 and InceptionV3 models to detect three types of BrTs: meningioma, glioma, and pituitary adenoma. They also used gradient-weighted classification activation maps (Grad-CAM) to explain the results and increase confidence. They evaluated the proposed model using a public dataset and achieved 99.02% accuracy, 98.75% precision, 98.98% recall, and an F1 score of 98.86%.

Pathak and Panika [82] investigated the early detection of BrTs using MRI images. They preprocessed the data for feature selection and noise reduction and utilized a CNN architecture for categorization. They evaluated the results using a 10-fold validation method. Their experiments showed that their proposed design achieved an optimal classification accuracy of 86.28% compared to existing classification methods. These results indicated that DL methods can significantly improve the detection of BrTs.

Nahiduzzaman et al. [83] investigated the accurate classification of four types of BrTs based on MRI images using a hybrid method of parallel sparse separation deep convolutional neural network (PDSCNN) and ridge regression complementary learning machine (RRELM). The results showed that the proposed model performed remarkably well with precision, recall, and average accuracy of 99.35%, 99.30%, and 99.22%, respectively, through five-fold validation, showing significant superiority compared to other models.

Table 6 aims to provide an overall comparison of different studies that have used DL techniques in the therapy and detection of BrTs. By highlighting the strengths and weaknesses of each approach, it provides a comprehensive picture of the trends and challenges in this field.

Table 6 Comparison of DL techniques for BrT therapy and detection

References	Objectives	Advantages	Drawbacks	DL techniques	Application	Evaluation metrics
[75]	Segment BrTs in MRI using three segmentation methods.	Uses three methods: high Dice for threshold-based segmentation.	Varying performance across methods; unclear best overall.	Threshold-based, region-based, CNN (U-Net) with pre-processing.	Tumor segmentation in MRI.	Accu, Prec, Rec, F1s
[76]	Classify BrTs into four categories using transfer learning and hyperparameter optimization.	Very high accuracy with CEQA; reduced training time.	Lacks details on precision, recall, and other metrics.	Pre-trained CNNs (EfficientNetB3, DenseNet121)+Co-Evolutionary Genetic Algorithm.	BrT classification.	Accu, Prec, Reca, F1s
[77]	Compare a custom CNN with pre-trained models using transfer learning.	VGG-16 outperformed others with high training and test accuracy.	Limited details on precision, recall, and F1-score.	CNN, AlexNet, and Inception-v3.	BrT classification.	Accu, Prec, Reca,
[78]	Classify BrTs using a mix of CNN, Swin Transformer, and EfficientNet.	Superior testing accuracy with Swin Transformer & EfficientNet; effective pre-processing.	It uses additional metrics (sensitivity, specificity) not on the standard list.	CNN, Swin Transformer, EfficientNet with normalization, resizing, and augmentation.	BrT classification.	Accu, Prec, Reca,
[79]	Detect brain cancer via DL and statistical feature extraction.	High accuracy is achieved with statistical operations on CNN-extracted features.	Heavy reliance on statistical classifiers; multiple data splits add complexity.	CNN (VGG-19, AlexNet) is used for feature extraction, and then KNN, RF, and SVM are used for classification.	Brain cancer detection from MRI.	Accu
[80]	Multi-class classification using DL combined with quantum genetic algorithms.	Robust and scalable; high reliability and accuracy.	Complex hybrid method.	Pre-trained EfficientNetB0+Quantum Genetic Algorithm for feature selection.	BrT classification.	Accu, Prec,
[81]	Develop an ensemble approach based on transfer learning for three tumor types.	High accuracy and explainability using Grad-CAM; improved trust.	Increased training complexity due to ensemble modifications.	Ensemble of DenseNet121 & InceptionV3 with modified classifiers and Grad-CAM.	BrT detection and classification.	Accu, Prec, Reca, F1s
[82]	Enable early diagnosis with a lightweight CNN.	Optimal accuracy with low computational overhead.	Achieves lower accuracy compared to other methods.	Lightweight CNN with thorough pre-processing and 10-fold cross-validation.	Early BrT classification.	Accu
[83]	Combine a lightweight PDSCNN with a hybrid RRELM for tumor classification.	Very high accuracy, precision, and recall; enhanced interpretability via SHAP.	Complex architecture; MCC not reported.	Lightweight parallel depthwise separable CNN (PDSCNN)+Hybrid Ridge Regression ELM; CLAHE & SHAP.	BrT classification.	Accu, Prec, Reca

5.3.1 Findings and Discussions

Data analysis in Table 6 shows that DL has been used in various studies to diagnose and classify BrTs with multiple approaches. Transfer Learning algorithms, CNNs, and hybrid algorithms such as Quantum Genetic Algorithm and Grad-CAM have played a key role in improving the accuracy of the models. Some methods, such as EfficientNet, DenseNet, and Swin Transformer, have performed better than traditional models. Existing challenges include the lack of accurate provision of evaluation criteria such as Prec, Reca, and F1s in some studies, the high complexity of hybrid methods, and the need for precise parameter tuning. Also, some methods, such as PDSCNN and RRELM, have

increased the interpretability of the models, which is of great importance in the reliability of medical results. Finally, the results show that using optimization methods such as Co-Evolutionary Genetic Algorithm and Whale Optimization can increase classification accuracy and improve the generalization ability of models.

5.4 Exploring Hybrid Techniques for BrT Identification and Treatment

In BrT detection and treatment, researchers recognize hybrid techniques that use SL or USL methods as powerful tools to improve the accuracy of diagnosis and prediction. These techniques focus on combining SL and USL algorithms

instead of using complex processes such as DL, RL, or FL. For example, researchers can use SVMs, an SL method, with USL algorithms like K-means clustering to identify and classify different BrT features. Additionally, specialists can use algorithms like K-Nearest Neighbors (KNN) and Decision Trees to classify and predict various types of BrTs in MRI (Magnetic Resonance Imaging) images. These hybrid approaches significantly improve the performance of diagnostic models by leveraging the power of both SL and USL techniques, helping to identify hidden patterns and features in medical data more accurately. This section will discuss articles that have tried to use hybrid techniques to identify and cure BrTs.

Bhimavarapu et al. [84] investigated the more accurate detection of BrTs using a fuzzy C-Means (FCM) segmentation algorithm, especially for MRI images. In this method, an additive learning machine (ELM) was used for classification, and the results obtained showed that the classification accuracy reached 98.56%, precision reached 99.14%, and recall reached 99.25%. The proposed model consistently outperformed other models and increased the classification accuracy by 1.21–6.23% compared to other models.

Ganapathy et al. [85] investigated the prediction of BrTs using a novel system based on K-Means and Mini-Batch K-Means clustering algorithms and CNN. The proposed method using these algorithms effectively performed the MRI image segmentation process and identified the location and presence of tumors. The results showed that the proposed system performed better than classical models such as logistic regression, support vector, K-nearest neighbor, random forest, and decision tree regarding prediction accuracy, precision, recall, F1 score, and prediction time.

Lokeswaran et al. [86] investigated the classification of BrTs using gene expression datasets. They used ML models such as decision tree (DT), SVM, logistic regression (LR), nearest neighbor (KNN), and random forest (RF) to classify different types of BrTs. The results showed that logistic regression outperformed other methods, such as random forest, SVM, decision tree, and KNN, identifying different kinds of BrTs.

Kumar et al. [87] investigated classifying BrTs into malignant and benign categories using ML and quantum computing. The QSVM model was executed on actual quantum machines and simulators to obtain performance metrics like accuracy and runtime. The results showed that the kernel-based QSVM model on a 32-qubit simulator was 188 times faster than the classical model and had an accuracy of 95%, 1.60% better than the classical model. Furthermore, the model on a 5-qubit superconducting processor with the same accuracy required 24.19% less time than the classical SVM model.

Alqhtani et al. [88] investigated the automatic segmentation of BrTs and a framework for classifying BrT regions. This method performed various steps, including preprocessing, enhancement of brain MRI images using CLAHE and diffusion filtering, and then segmentation of the target region using the Fuzzy C-Means (FCM) clustering technique. The results showed that this method significantly improved image contrast. In performance evaluation, sensitivity was 0.977, specificity was 0.979, accuracy was 0.982, and Dice score (DSC) was 0.961.

Guerroudji et al. [89] investigated the effective detection of brain cancer using ML technology. The performance of the Bayesian network compared with other classification algorithms such as SVM, MLP, KNN, Random Forest, Decision Tree, XGBoost, LGBM, Gaussian Process, and RBF SVM showed that the Bayesian network is superior in this field. The experimental results showed that this method has 100% sensitivity and more than 98% classification accuracy for tumors, demonstrating the high effectiveness of cancer cell segmentation.

Jareena and Chokkalingam [90] investigated a method with a high probability of success in detecting BrTs. They implemented this approach using an image segmentation technique and a classifier, using the Random Forest algorithm as the classifier. They tested this method using a dataset on the Kaggle platform, and their results showed an accuracy of approximately 97.68%. They found that the proposed system was superior in detection rate, false positive rate, and recall compared to other methods, demonstrating a high ability to detect BrTs.

Khekare et al. [91] investigated using ML techniques to detect and classify BrTs in magnetic resonance imaging (MRI) images. They studied different ML algorithms, preprocessed the MRI data, built and trained models, and evaluated their performance. The research results emphasized the potential of AI-based methods in medical image analysis, especially for detecting and classifying BrTs.

Hossain et al. [92] investigated the detection of BrTs using brain MRI images. In this regard, a preliminary processing method for brain MRI images used Otsu thresholding and morphological operations. Then, various ML models, including CNNs and nearest neighbors (KNN), were used for classification. The results of this study showed that the combined CKNN model performed very well in detecting BrTs, with an accuracy of 89.88%.

Basthikodi et al. [93] investigated the improvement of multiclass classification of BrTs using the SVM algorithm and feature extraction techniques such as histogram of oriented gradients (HOG) and local binary pattern (LBP). In this study, they achieved an accuracy of 95.95% by combining HOG and LBP feature extraction techniques with SVM. Finally, the combination of SVM with HOG, LBP,

and PCA produced the highest performance improvement and achieved an accuracy of 96.03%, F1 score of 96.00%, precision of 96.02%, and recall of 96.03%.

Subramaniam and Krishnasamy [94] presented an ABES model for BrT detection. They used a Bi-GRU layer to learn crucial contextual information and an attention layer to assign weights to the outputs of the Bi-GRU layer. Then, they performed tumor classification using a clustered SVM

classifier. They trained this model using an improved whale optimization algorithm to optimize the weights of the ABES classifier. The results showed that the ABES model achieved 97.2% accuracy, 98.61% precision, 96.48% specificity, and 97.34% sensitivity, indicating its excellent performance in BrT classification.

Table 7 aims to provide an overall comparison of studies that have used Hybrid techniques in the therapy and

Table 7 Comparison of hybrid techniques for BrT therapy and detection

References	Objectives	Advantages	Drawbacks	DL techniques	Application	Evaluation metrics
[84]	Improve segmentation and classification using Fuzzy C-Means and Extreme Learning Machine (ELM).	Higher accuracy than existing models; robust classification.	Computational complexity; artifacts affect accuracy.	Improved Fuzzy C-Means clustering + Extreme Learning Machine (ELM).	BrT segmentation and classification.	Accu, Prec, Reca
[85]	Develop a hybrid prediction system using segmentation and CNN.	Clear tumor identification; better accuracy than existing classifiers.	High computation time requires post-processing for clarity.	K-Means & Mini-Batch K-Means + Morphological operations + CNN.	BrT prediction and segmentation.	Accu, Prec, Reca, F1s
[86]	Classify BrTs using gene expression data.	Identifies multiple tumor types; improves early detection.	Small dataset (40 patients); limited generalizability.	Principal Component Analysis (PCA) + Pearson correlation + ML classifiers (LR, SVM, DT, RF, KNN).	BrT classification based on gene data.	Accu, Prec, Reca,
[87]	Use quantum ML (QML) for BrT classification.	Faster processing; better accuracy than classical SVM.	Quantum hardware limitations require a specialized setup.	Quantum SVM (QSVM).	Classifying BrTs into benign/malignant.	Accu
[88]	Develop an automated segmentation and classification system.	Improved contrast and classification efficiency.	Processing time is still a challenge.	CLAHE + Diffusion filtering + Fuzzy C-Means + SVM.	BrT segmentation and classification.	Accu
[89]	Enhance BrT detection using Bayesian networks.	High sensitivity and accuracy; robust segmentation.	Dependence on feature extraction quality.	Gradient Vector Flow (GVF) Snake + Bayesian network + Morphological features.	BrT segmentation and classification.	Accu, Prec
[90]	Improve BrT detection using PCA and DWT with RF.	High detection accuracy; reduced false positives.	Requires manual parameter tuning.	Principal Component Analysis (PCA) + Discrete Wavelet Transform (DWT) + Random Forest.	BrT detection and classification.	Accu
[91]	Utilize ML for MRI-based tumor detection and classification.	AI-driven method improves diagnosis accuracy.	Requires large datasets for robustness.	Custom CNN + Pre-trained models + ML classifiers.	BrT detection and classification.	Accu
[92]	Enhance MRI-based tumor segmentation with CKNN.	Improved classification performance.	Dataset size limitations.	Otsu's Thresholding + Morphological Operations + CKNN (CNN + KNN).	BrT segmentation and classification.	Accu, Prec, Reca,
[93]	Improve multi-class tumor classification using feature extraction and SVM.	Strong classification performance (96.03%).	Computational complexity.	HOG + LBP + PCA + SVM.	Multi-class BrT classification.	Accu, Prec, Reca, F1s
[94]	Develop an ensemble model combining Bi-GRU and SVM.	High classification accuracy; explainability via attention mechanism.	Requires tuning of multiple components.	Attention Bi-GRU + Ensemble SVM + Whale Optimization Algorithm.	BrT detection and classification.	Accu, Prec, Reca

detection of BrTs. By highlighting the strengths and weaknesses of each approach, it provides a comprehensive picture of the trends and challenges in this field.

5.4.1 Findings and Discussions

The data analysis in Table 7 shows that researchers have used various hybrid techniques to detect and classify BrTs, each offering specific advantages and disadvantages. In the segmentation section, researchers applied methods such as fuzzy C-means, K-means clustering, and Otsu's thresholding, achieving high accuracy. However, these methods present challenges as their computational complexity increases, making them more noise-resistant. In the classification section, methods such as Extreme Learning Machine (ELM), SVM, Random Forest (RF), and Bayesian Networks have performed well in detecting and classifying tumors. However, some of these methods require manual parameter tuning. Quantum SVM (QSVM) is faster than classical SVM but has hardware limitations. Some methods, such as Attention Bi-GRU+Ensemble SVM and HOG+LBP+PCA+SVM, have achieved high accuracy (over 96%), but the high complexity of the model is their main challenge. Also, methods such as Principal Component Analysis (PCA) and Discrete Wavelet Transform (DWT) have helped reduce the dimensionality of the data, but their parameter tuning is challenging. Combining image processing methods, ML, and optimization techniques in hybrid models has generally improved access and efficiency. However, robust processing and fine-tuning of parameters remain challenging in this field.

Fig. 8 Comparative analysis of the objectives of ML over methods in BrT therapy and detection

6 Discussions

This section examines the various factors used in evaluating studies related to the application of ML in the therapy and detection of BrTs. This analysis helps to understand and explain the role and impact of these factors in various aspects of the therapy and detection process. Evaluating these factors identifies the strengths and limitations of each method reviewed. It helps to determine their most appropriate application in different areas of medical science, especially in BrTs.

- **PQ1:** What are the primary objectives of using ML in BrT therapy and detection?

Figure 8 shows a comparative analysis of the objectives for using ML techniques to identify and cure BrTs based on the proposed classification. The proposed classification includes six main categories: Improving Model Accuracy and Performance (IMAP), Enhancing Data Privacy and Security (EDPS), Optimizing Computational Efficiency (OCE), Enhancing Model Generalization and Adaptability (EMGA), Improving Model Interpretability and Transparency (IMIT), and Other (OTH). Based on the analysis of this classification, the percentage of objectives related to IMAP, EDPS, OCE, EMGA, IMIT, and OTH are 35%, 21%, 18%, 12%, 8%, and 6%, respectively. This diagram depicts that most goals are related to IMAP, accounting for 35% of the identified goals.

- **PQ2:** What are the main advantages of ML compared to traditional methods in BrT therapy and detection?

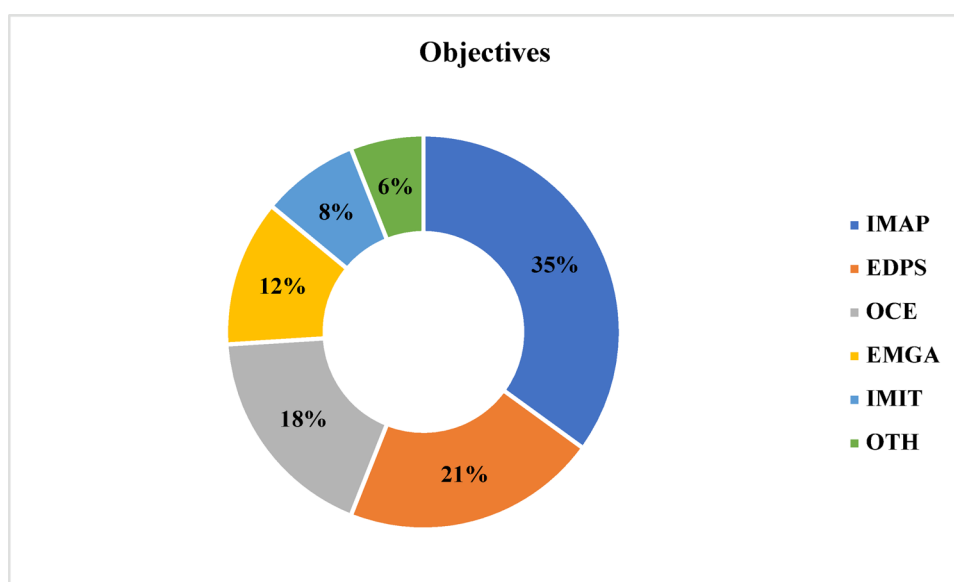


Fig. 9 Comparative analysis of the advantages of ML over traditional methods in BrT therapy and detection

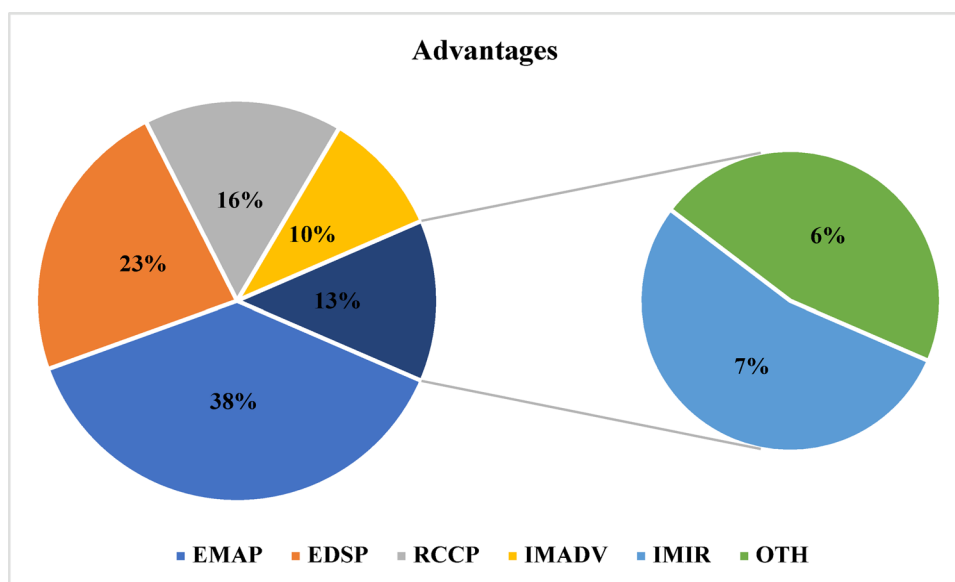


Fig. 10 Comparative analysis of drawbacks in using ML for BrT therapy and detection

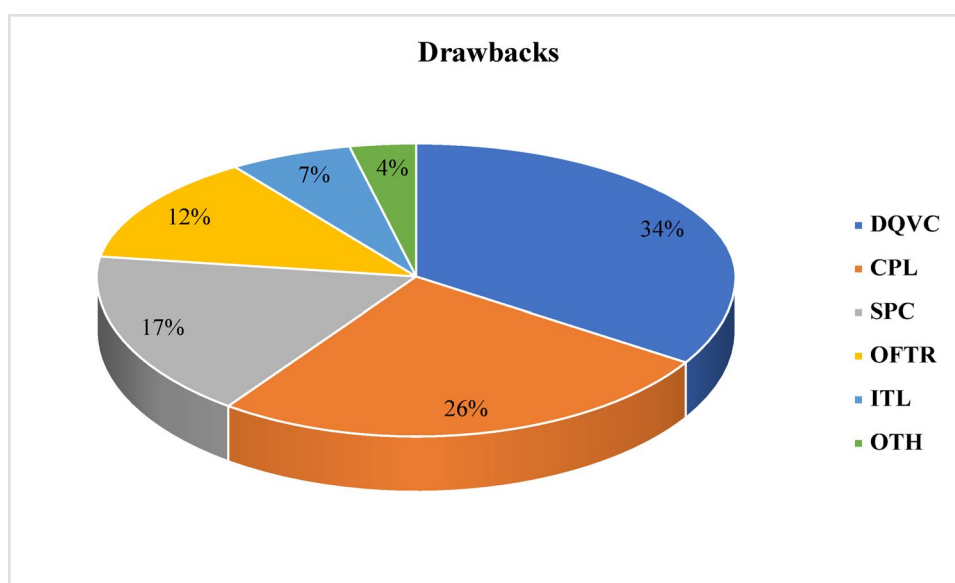


Figure 9 displays a comparative evaluation of the benefits of using ML strategies in identifying and curing BrTs according to the suggested classification. The suggested classification consists of six primary categories: Enhancing Model Accuracy and Performance (EMAP), Ensuring Data Security and Privacy (EDSP), Reducing Computational Costs and Processing Time (RCCP), Improving Model Adaptability to Data Variability (IMADV), Improving Model Interpretability and Reliability (IMIR), and OTH. Based on the evaluation of this classification, the proportion of advantages associated with EMAP, EDSP, RCCP, IMADV, IMIR, and OTH are 38%, 23%, 16%, 10%, 7%, and 6%, respectively. This diagram depicts that most of the identified benefits are related to EMAP, which accounts for 38% of the identified benefits.

- **PQ3:** What are the major drawbacks of using ML techniques for BrT therapy and detection?

Figure 10 displays a comparative evaluation of the drawbacks identified using ML techniques to identify and cure BrTs according to the suggested classification. The suggested classification consists of six primary categories: Data Quality and Variability Challenges (DQVC), Computational and Processing Limitations (CPL), Security and Privacy Constraints (SPC), Optimization and Fine-Tuning Requirements (OFTR), Interpretability and Transparency Limitations (ITL), and Other (OTH). Based on the evaluation of this classification, the proportion of drawbacks associated with DQVC, CPL, SPC, OFTR, ITL, and OTH is 34%, 26%, 17%, 12%, 7%, and 4%, respectively. This

Fig. 11 Comparative analysis of ML applications in BrT therapy and detection

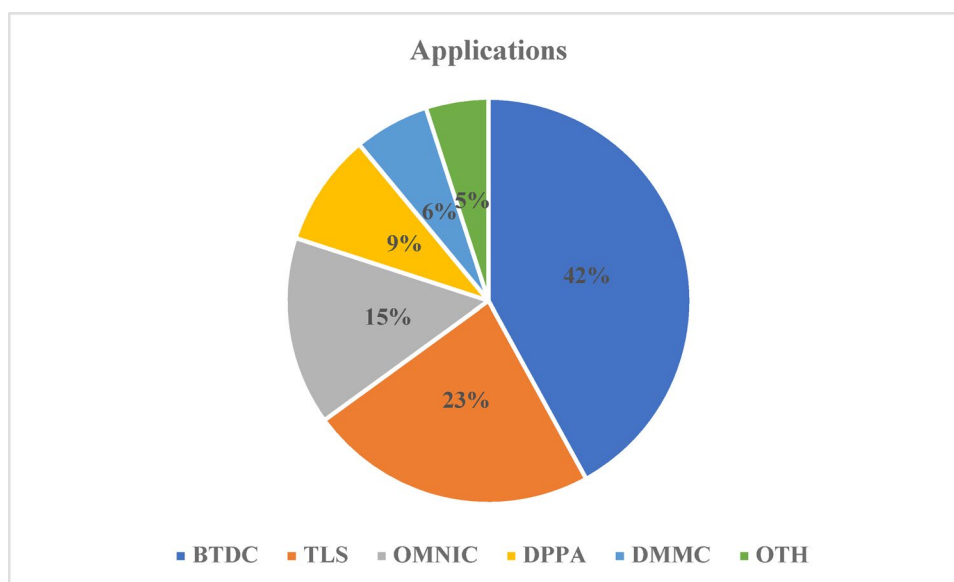


Fig. 12 Comparative analysis of evaluation metrics in ML models for BrT therapy and detection

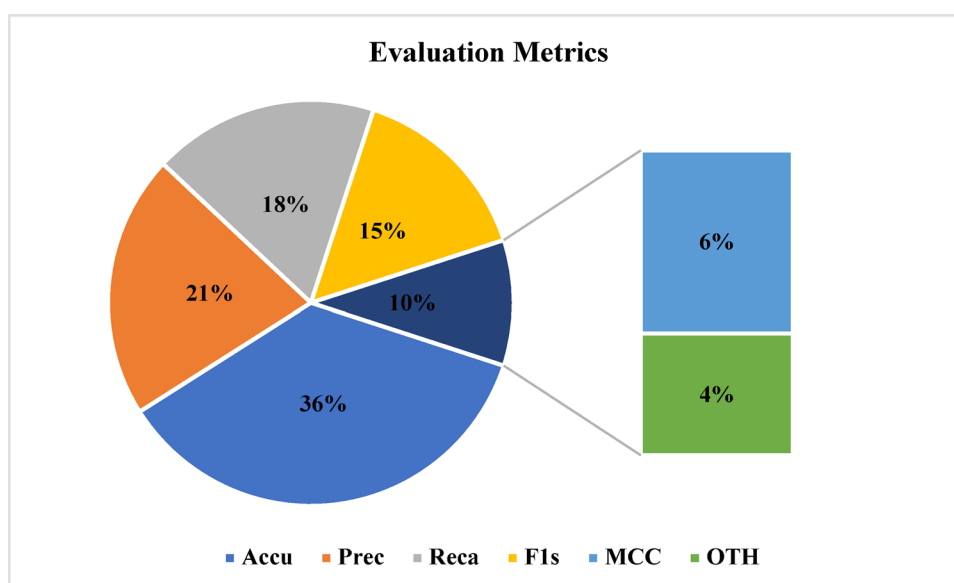


diagram depicts that most of the identified drawbacks relate to DQVC, accounting for 34%.

- **PQ4:** What are the applications of ML for BrT therapy and detection?

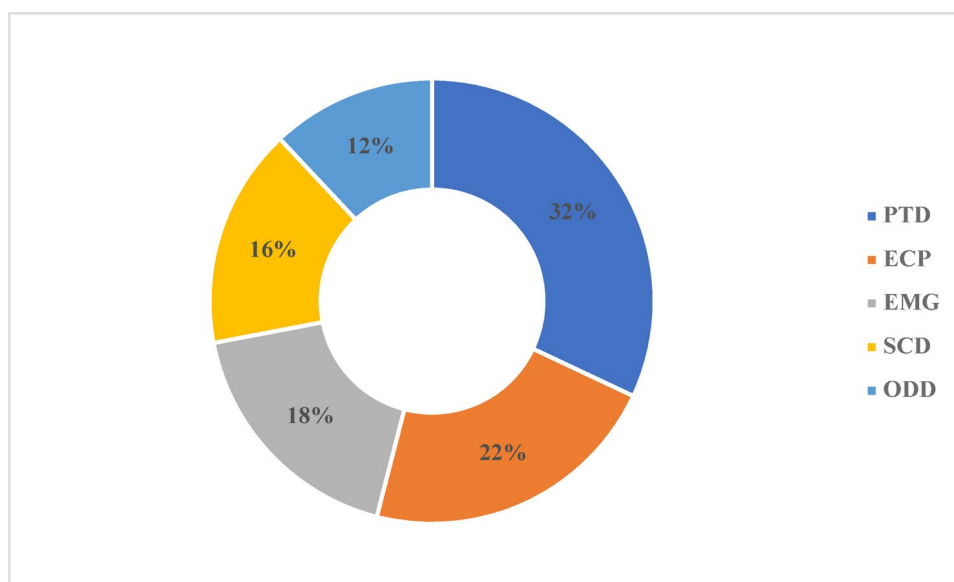
Figure 11 displays a comparative evaluation of the applications identified for using ML techniques to identify and cure BrTs according to the suggested classification. The suggested classification consists of six primary categories: BrT Diagnosis and Classification (BTDC), Tumor Lesion Segmentation (TLS), Optimization of ML Models for Non-Ideal Conditions (OMNIC), Disease Progression Prediction and Assessment (DPPA), Data Management and Multi-Center Collaboration (DMMC), and Other (OTH).

Based on the evaluation of this classification, the proportion of applications associated with BTDC, TLS, OMNIC, DPPA, DMMC, and OTH are 42%, 23%, 15%, 9%, 6%, and 5%, respectively. This diagram depicts that most identified applications are related to BTDC, accounting for 42% of the total applications.

- **PQ5:** What are the key evaluation metrics for assessing ML models in BrT therapy and detection?

Figure 12 displays a comparative evaluation of the evaluation metrics used in ML models for identifying and curing BrTs according to the suggested classification. The suggested classification consists of six primary categories: Accu, Prec, Reca, F1s, MCC, and Other (OTH). Based on

Fig. 13 Comparative analysis of Key ML parameters for BrT therapy and detection



the evaluation of this classification, the proportion of evaluation metrics associated with Accu, Prec, Reca, F1s, MCC, and OTH is 36%, 21%, 18%, 15%, 6%, and 4%, respectively. This diagram depicts that most of the evaluation criteria used are related to Accu, accounting for 36% of the identified evaluation metrics.

Figure 13 displays a comparative evaluation of key parameters used in ML systems for BrT therapy and detection based on the proposed classification. Based on the study of this classification, the percentage of parameters related to Precise Tumor Detection (PTD), Efficient Computational Processing (ECP), Enhanced Model Generalization (EMG), Seamless Clinical Deployment (SCD), and Optimized Data Dependency (ODD) are 32%, 22%, 18%, 16%, and 12%, respectively. This figure shows that improving PTD and ECP are the main priorities in developing ML systems for BrT therapy and detection.

7 Open Issues

Researchers and medical professionals increasingly use AI and ML in the medical field, especially for identifying and curing complex diseases such as BrTs. These technologies enhance the accuracy and speed of diagnosis and assist doctors in making clinical decisions. However, several obstacles and challenges hinder their application in the real world, affecting their performance, reliability, and acceptance. The following section will discuss some of the most important of these open issues.

- **Data Imbalance in Medical Imaging** One of the main challenges in applying ML to BrT diagnosis is data imbalance, where some rare tumor types are less

represented in the dataset. Hence, this leads to model bias and reduces diagnostic accuracy in less common cases. To address this problem, methods such as data augmentation (using GANs, rotation, and inversion of images), oversampling and under-sampling, and cost-sensitive learning are used. Also, few-shot and self-supervised learning allows for better generalization of models in limited data conditions [95].

- **Model Generalizability and Domain Adaptation** ML models trained on specific datasets often perform poorly in real-world clinical situations. This is due to differences in imaging protocols, scanner types, and patient demographics. Transfer learning is used to adapt models to new data to improve generalizability. Also, domain adaptation methods such as adversarial and FL help reduce the differences between different data distributions. These approaches enhance the robustness of models across diverse clinical environments [96].
- **Explainability and Transparency in Medical AI** DL models are often considered black boxes because the reasons for their decisions are not clear. This lack of transparency reduces clinicians' confidence in the model output. Hence, to address this problem, XAI methods such as Grad-CAM (highlighting important image areas), SHAP (measurement of the impact of features), and LIME (explaining model behavior with simpler models) are used. These methods increase model transparency and facilitate its clinical adoption [97].
- **Optimization of the Learning and Decision-Making Process** The process of learning models requires a large amount of labeled data, which is time-consuming and costly in the medical environment due to the specialization of the information. High dependence on labeled data can cause problems such as overfitting and reduced

performance in low-data conditions. Also, the clinical decision-making process is a significant challenge in using intelligent models due to the individual complexities of patients and the variability of treatment outcomes. These challenges require developing methods to reduce dependence on labeled data and improve decision-making [98].

- **Security and Privacy of Medical Data** Maintaining the protection and privacy of sensitive patient data, especially in multicenter environments, is one of the critical concerns in using modern technologies. The transmission and processing of medical data are faced with risks such as inference attacks and unauthorized access. Legal restrictions and privacy constraints make it difficult to use sensitive data freely. Hence, this is a serious challenge when using FL methods [99].
- **Standardization and Evaluation of Model Performance** One fundamental problem in using ML models in medicine is the lack of standardized and comprehensive methods for evaluating their performance. Many models are tested in controlled environments and with specific data, while their performance in real-world conditions may suffer from reduced accuracy. The lack of longitudinal studies and independent evaluations makes it difficult to determine the exact effectiveness of models [100].

7.1 Suggestions Enhancements for BrT Therapy and Detection Using ML Techniques

In recent years, significant advances in ML and DL have played an essential role in treating and detecting BrTs. However, the application of these models is accompanied by several challenges. The following attempts to answer the primary question 6 (PQ6) by providing suggestions for the existing challenges.

- **PQ6:** What strategies can enhance the performance of ML models in BrT therapy and detection?
- **Improving Data Quality and Increasing Datasets** One of the most critical challenges for ML models in identifying and curing BrTs is the quality and volume of training data. Medical data, especially Magnetic Resonance Imaging (MRI) and Computed Tomography (CT) images, usually have noise, variable resolution, and complex features that can affect the performance of DL models. Therefore, preprocessing methods such as Bias Field Correction, Histogram Equalization, Denoising Filters, and Normalization can help improve data quality. Increasing the volume of training data is also necessary to enhance the generalizability of models. In many cases, existing datasets have a limited number of

labeled examples, which can lead to overfitting of ML models. One of the key solutions to this problem is data augmentation. Techniques such as Geometric Transformations (rotation, scaling, angle change), Intensity Transformations (change contrast and brightness), and GAN-based Augmentation (generate artificial data using Generative Adversarial Networks (GANs)) can help increase the diversity of training data. In addition, integrating multi-center data from different hospitals and research centers through FL can improve ML models' performance. This approach allows medical centers to train models on more diverse datasets without directly sharing sensitive patient data. Hence, this increases the accuracy of diagnostic models and reduces bias. However, researchers and developers must properly manage challenges such as Domain Shift, Data Heterogeneity, and Privacy Constraints to utilize FL effectively. Finally, increasing data quality through Label Refinement and reducing errors due to Interobserver Variability among radiologists and pathologists can significantly impact model performance. Consensus Labeling Techniques, Active Learning, and Weakly Supervised Learning can help improve the labeling process and increase the accuracy of ML models in diagnosing and classifying BrTs [101–103].

- **Optimization of ML Models** Optimizing ML models for identifying and curing BrTs requires developing advanced techniques and combining different models. One of the key methods in this field is the combination of DL and RL, which can dynamically and adaptively improve the therapy and detection process. For example, Deep Q-Networks (DQN) and CNN models can enable more accurate identification of tumor-related features. In addition, modern architectures such as Transformers, which were initially developed for Natural Language Processing (NLP), are now also used in Medical Imaging. Models such as Vision Transformers (ViT) and Swin Transformers perform better in extracting complex features from medical images. Also, Graph Neural Networks (GNNs) can model the complex connections between cancer cells and adjacent tissues, leading to more accurate diagnosis. On the other hand, reducing computational resource consumption is an important challenge in implementing these models in clinical environments. The use of Model Compression techniques such as Weight Pruning (removing unnecessary weights in the neural network), Quantization (reducing the precision of numerical calculations to reduce memory consumption), and Knowledge Distillation (transferring knowledge from large models to smaller models) can help to more efficiently implement these models in Edge Devices such as medical imaging systems [104–106].

- **Improving Model Interpretability** One of the significant challenges in using ML in medicine is the interpretability of learning models' decisions. Clinicians need to understand why and how a model reached a particular conclusion. The use of Explainable AI is essential to solve this challenge. Methods such as Layer-wise Relevance Propagation, Shapley Additive explanations, and Local Interpretable Model-agnostic Explanations can explain which parts of the image or patient data impacted the model's decision most. In addition, combining Supervised and Unsupervised learning in Semi-supervised Learning models can lead to improved model performance. This approach can use Unlabeled information to extract meaningful features, especially when labeled data is limited. Also, integrating multimodal data from various sources, including Medical Imaging, Genomic Data, Histopathological Data, and Electronic Health Records, into the Multimodal Learning framework can lead to a more comprehensive view of the patient's condition [107, 108].
- **Improving the Learning and Decision-Making Process** One of the efficient methods to reduce the need for labeled data and optimize the learning process is to use Active Learning. In this method, the model can select the most ambiguous examples for labeling, which reduces manual labeling costs and increases the model's efficiency. In addition, using RL in the treatment process can help with Personalized Treatment Plans. For example, a RL agent can choose the best option for each patient by simulating the possible outcomes of different treatment methods. This method, along with Markov Decision Processes (MDPs) and Partially Observable MDPs (POMDPs), can be very effective in clinical decision-making. On the other hand, Transfer Learning allows the reuse of models trained on similar diseases to diagnose BrTs. This technique reduces the need for large data sets and increases models' accuracy in data-poor environments [109, 110].
- **Improving Security and Privacy in FL Models** FL is an approach that allows healthcare facilities to improve ML models without sharing patient data directly. However, this approach still faces security challenges. One key solution to increase security is using Homomorphic Encryption, which allows calculations to be performed on encrypted data without decryption. In addition, Personalized FL allows each healthcare facility to tailor learning models to the characteristics of its patients without sharing raw data with other facilities. This approach can use meta-learning techniques to optimize the process of customizing models. Another advanced approach to preserving privacy is Differential Privacy, which reduces the possibility of identifying

individual information by adding noise to the data. This technique, combined with Secure Multi-Party Computation (SMPC), can ensure the security of patient data while processing FL models [111, 112].

- **Developing Comprehensive and Standard Evaluation Methods for ML Models** One of the fundamental issues in using ML for identifying and curing BrTs is the lack of standardized and comprehensive evaluation methods. Many DL and Supervised Learning models used for medical image analysis are tested in controlled environments and with specific datasets. However, their performance in real-world conditions (with diverse and noisy data) may suffer from significant degradation. Hence, to solve this problem, external validation on independent and multicenter datasets is needed to examine the performance of models under different clinical conditions. In addition, using Longitudinal Studies is essential to investigate the impact of ML models in improving the treatment process. Combining traditional evaluation methods, such as the Receiver Operating Characteristic (ROC) Curve, Dice Similarity Coefficient (DSC), and Intersection over Union (IoU), with newer metrics, such as Causal Inference-Based Metrics and Survival Analysis Models, can provide a better view of the effectiveness of ML models [113, 114].

7.2 Future Directions in Using ML for BrT Therapy and Detection

Recent advances in AI and ML are transforming how we identify and cure BrTs. These technologies can increase the accuracy and quick diagnosis, but issues still need to be addressed to optimize these processes. Researchers are looking for innovative strategies to enhance the efficiency of these systems. Below, we will review some of the most important future directions in this field that can optimize the identification and curing of BrTs.

- **Integrating AI with Quantum Computing** One of the main limitations of ML models in medical data processing is their need for high computing power and long processing time. Quantum Computing can solve problems related to computational complexity by providing new processing methods. Quantum ML (QML) algorithms can analyze medical imaging data much faster using quantum overlap and entanglement. Also, quantum models can better identify complex patterns in genetic and clinical data and increase the accuracy of diagnosing and predicting tumor progression. On the other hand, using quantum optimization algorithms such as Quantum Approximate Optimization Algorithm (QAOA) can lead to improved feature selection

processes and reduced dimensionality of medical data. These advances could advance the diagnosis of BrTs to a higher level of accuracy and speed, ultimately leading to improved quality of treatment.

- Advanced FL for Multicenter Data Exploitation** One of the significant challenges in ML models in medicine is limited access to diverse data from different centers. Due to patient privacy issues, the transfer of medical data between hospitals faces legal restrictions. FL is a novel approach that allows hospitals and research centers to train collaborative ML models without directly sharing raw data. The data remains local in this approach, and only model updates are shared with other centers. Using Homomorphic Encryption and Secure Multiparty Computation (SMPC) in FL allows models to improve without exposing sensitive patient information. Also, the development of personalized FL will enable models to be tailored specifically for each hospital or patient, leading to higher accuracy in therapy and detection.
- Multimodal Models for Simultaneous Data Analysis** Accurate diagnosis of BrTs requires analyzing diverse data, including medical imaging, genetic data, pathology, and electronic health records (EHRs). Multimodal learning models can process this heterogeneous data simultaneously and extract valuable information. New architectures such as Multimodal Transformers and Graph Neural Networks (GNNs) allow AI to understand the complex relationships between different data better. For example, these models can combine information from genetic tests with medical imaging results to provide more accurate diagnoses. Also, self-supervised learning can enable models to identify essential features without extensive labeled data. These methods will lead to reduced diagnostic errors and better personalized treatment.
- Self-Supervised Learning (SSL) to Address the Lack of Labeled Data** One of the key challenges in using deep learning models for BrT detection is the need for large amounts of labeled medical data, which is expensive and time-consuming to collect. As a novel approach, self-supervised learning (SSL) enables learning important features from unlabeled data through pretext tasks and then training the model with a limited amount of labeled data. Effective techniques in this area include SimCLR, BYOL (Bootstrap Your Own Latent), and contrastive learning, which improve the detection accuracy of ML models by extracting rich representations from MRI images without the need for large amounts of labeled data. By creating stable latent maps, these methods enable the model to recognize complex pathological patterns better and increase its robustness to changes in the input data. Developing and optimizing these techniques can

significantly reduce the challenges associated with the lack of medical data and improve the performance of diagnostic models in clinical scenarios.

- Integrating Radiomics Features into ML Models** Radiomics, as a novel approach in medical image analysis, allows the extraction of quantitative features from MRI and CT images that can effectively improve the diagnosis and prediction of tumor behavior. These features include pixel intensity statistics, texture, shape, and wavelet transform-based features that can reveal hidden information not visible to the human eye. Combining radiomics features with deep learning models (such as CNNs and Transformers) and classical methods, such as SVM and random forests (RF), can improve the accuracy and interpretability of diagnostic models. One of the key challenges in this area is selecting optimal features and reducing the dimensionality of the extracted information, which can be optimized using supervised feature selection and principal component analysis (PCA) algorithms. Future research should focus on integrating radiomics with multimodal learning models and investigating its impact on the accuracy and sensitivity of BrT diagnosis to help increase the reliability and efficiency of AI systems in clinical settings.

7.3 Limitations of the Study

One of the main limitations of this review is the reliance on previous studies, each of which has significant differences in methodology, sampling, and reporting methods. This heterogeneity in study design and statistical methods can introduce systematic biases and reduce the generalizability of the results to diverse clinical populations. In addition, the time constraints caused by the publication dates of the reviewed articles prevent the findings from older studies from being quickly adapted to recent advances in ML techniques and new technologies. Also, the lack of empirical data and practical evaluations of the proposed methods limits the possibility of directly validating the performance of these methods in clinical settings. Similarly, the lack of unified standards for evaluating and comparing results between studies is a serious challenge in extracting comprehensive and definitive conclusions, which requires the development of common frameworks for assessing the performance of algorithms in future studies.

8 Conclusion

BrTs can be primary or metastatic in origin and continue to grow uncontrollably under genetic factors, epigenetic changes, cellular mutations, and aberrant signaling in

molecular. In addition, these tumors challenge the blood-brain barrier and the structural complexity of the brain, leading to a decrease in the effectiveness of standard therapeutic methods such as surgery, radiotherapy, and chemotherapy. For this reason, developing new diagnostic and therapeutic strategies based on advanced technologies and AI is essential. ML facilitates the analysis of large volumes of medical data using complex algorithms. It allows the extraction of hidden patterns for more accurate diagnosis and optimization of treatment protocols. Using deep neural networks, RL models, transfer learning, and hybrid methods has improved accuracy in automatic tumor diagnosis, disease progression prediction, and treatment personalization based on individual patient characteristics. This paper systematically reviews the techniques of DL, FL, RL, and hybrid models used in identifying and curing BrTs by providing a detailed taxonomy and a comprehensive classification of these technologies is provided. The analysis shows that developing ML systems for BrT management should improve PTD by 32%, ECP by 22%, EMG by 18%, SCD by 16%, and ODD by 12%. Among these, more accurate tumor diagnosis and optimization of computational processing are identified as the most important priorities for future research. This study's findings show that using ML improves the accuracy of radiological diagnosis, reduces dependence on human analysis, increases the speed of clinical data processing, and optimizes medical decision-making methods. Also, by combining multimodal data such as genetic information, clinical history, and medical images, this technology can personalize each patient's treatment path and improve survival rates and quality of life for patients with BrTs. Finally, by carefully assessing the existing issues and presenting scientific solutions to overcome them, this study can be a comprehensive guide for researchers and experts to develop advanced AI models and improve BrT therapy and detection systems.

For future research, we can focus on developing ML models that can cope with the heterogeneity of medical data and the limitations of training data. One potential path is to use FL and transfer learning to improve the generalizability of models in different clinical settings. In addition, the development of explainable AI algorithms can help increase the acceptance of these methods in the medical community. Also, investigating the combination of classical image processing methods with deep learning techniques can help improve the accuracy of diagnosis. Finally, conducting experimental studies in real clinical settings with medical experts' participation can help evaluate the practicality and optimization of the proposed methods.

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Declarations

Competing Interests The authors declare no competing interests.

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